Chapter 4

Peak-Current-Mode Control

Outline

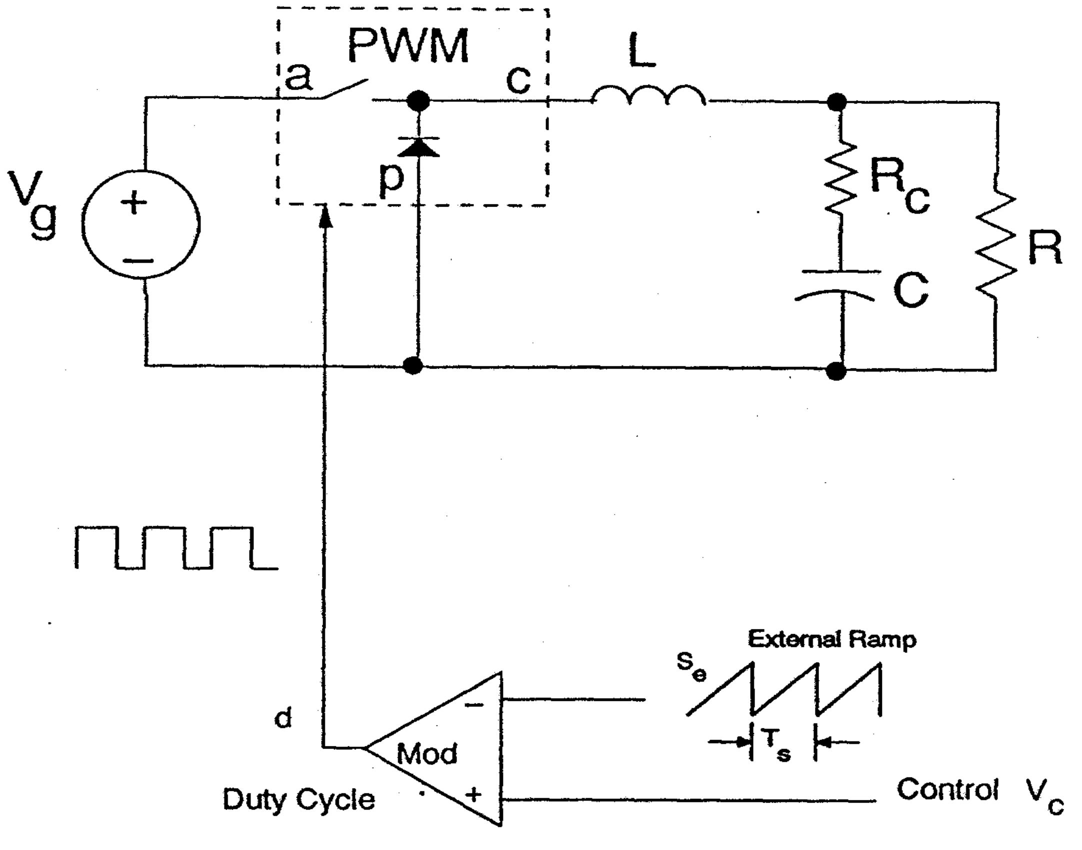
- **Definition of current-mode control**
  - **Hysteretic current-mode Control**
  - **'SCM' Control** standard control module
  - **'CIC' Control** current injection control
  - **'Average Current' Control**

- **Benefits and Characteristics of Current-Mode Control**
  - **Current Limiting**
  - **Current Sharing**
  - **Simple Compensation**

- **Issue: Current-loop Instability**

Before Current-Mode Control…

Single-Loop/Voltage-Mode Control



(a) Buck

- **Duty cycle is generated by clock, ramp signal, and error voltage**

History of Current-Mode Control

Hysteretic current-mode Control---1967

L.E. Gallaher, “Current Regulator with AC and DC feedback,” U.S. Patent 3,350,628, 1967.

‘SCM’ Control---1972

F.C. Schwarz, “Analog Signal to Discrete Time Interval Converter (ASDTIC),” U.S. Patent 3,659,184, 1972

A.D. Schoenfeld, Y. Yu, “ASDTIC Control and Standardized Interface Circuits Applied to Buck, Parallel and Buck-Boost DC-to-DC Power Converters,” NASA Report NASA CR-121106, Prepared by TRW Systems, February, 1973.

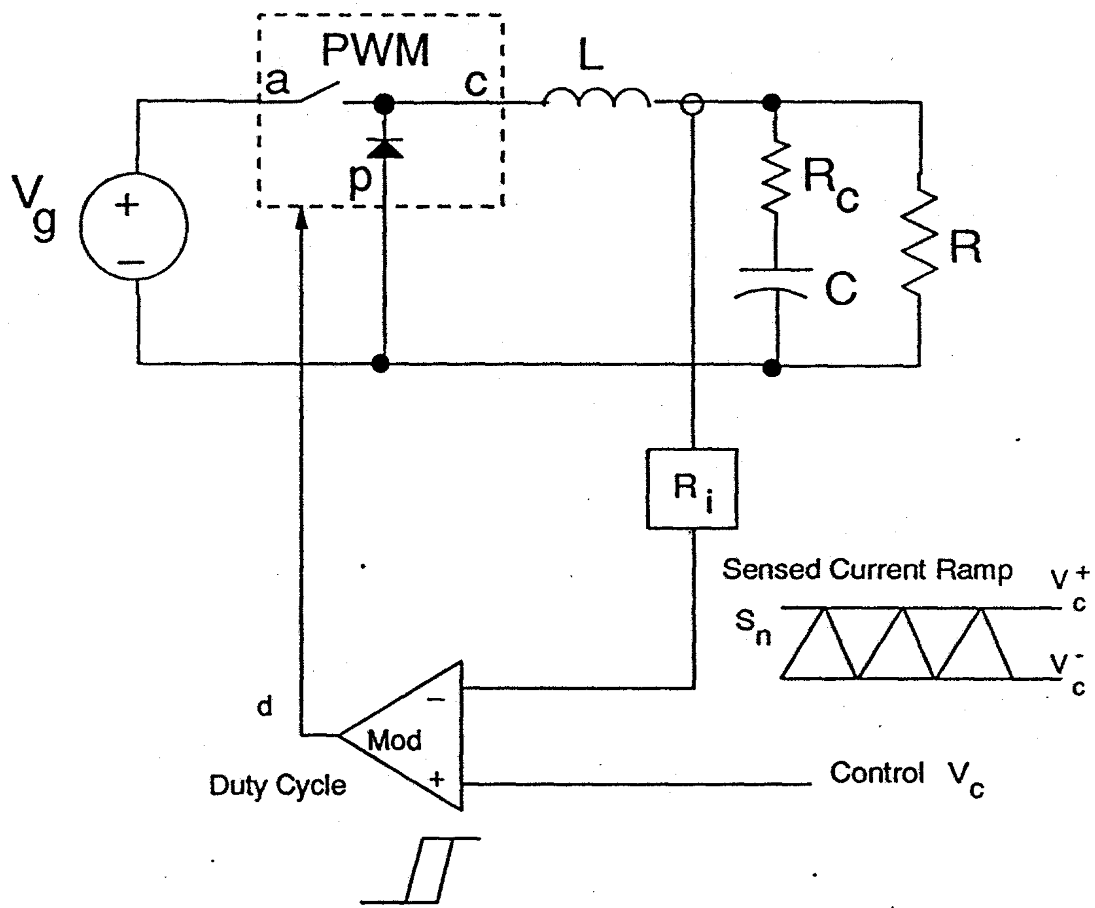
‘CIC’ Control (peak current mode control) –1978

C.W. Deisch, “Switching Control Method Changes Power Converter into a Current Source” IEEE Power Electronics Specialists Conference, 1978 Record, pp.300-306.

‘Average Current’ Control--1977

P.L. Hunter, “Converter Circuit and Method Having Fast Responding Current Balance and Limiting,” U.S. Patent 4,002,963, 1977.

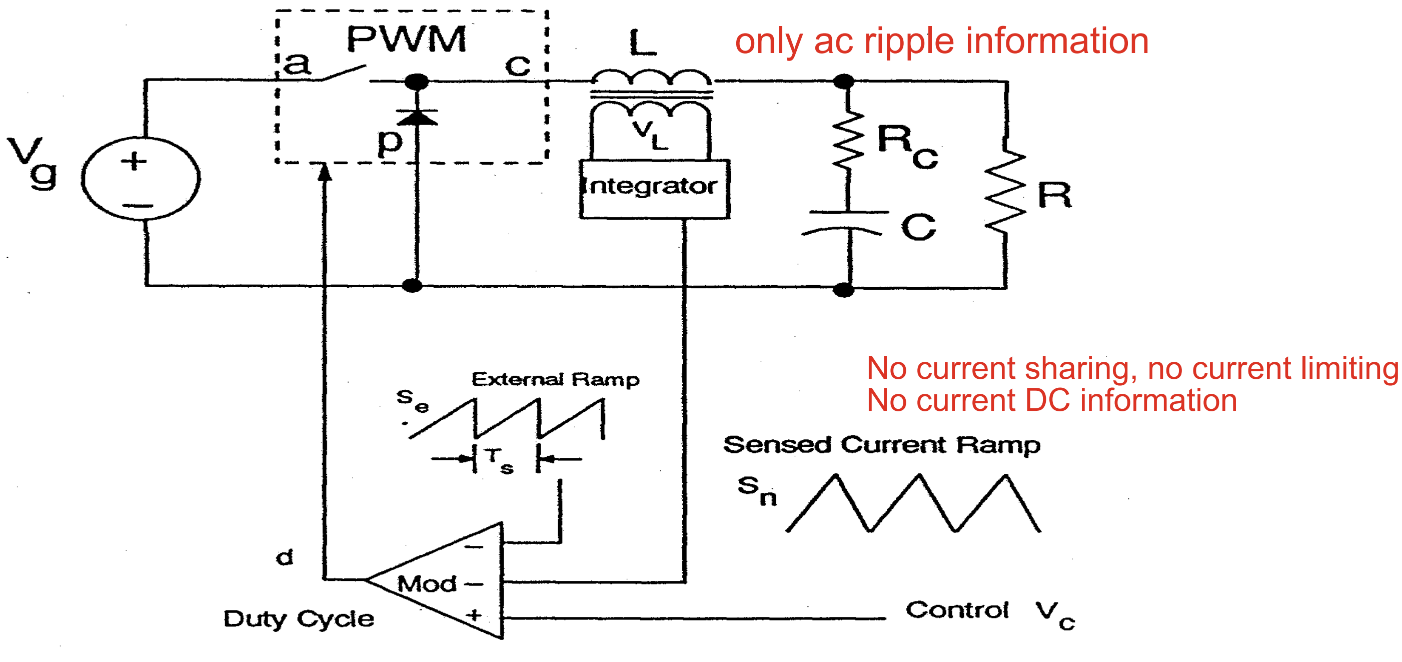
Hysteretic current-mode control



- **Current waveform is used to determine both the turn-on and turn-off times**

Definition of Current-Mode Control

Original implementation of current-mode control (1972) "Standard Control Module" (SCM) implementation



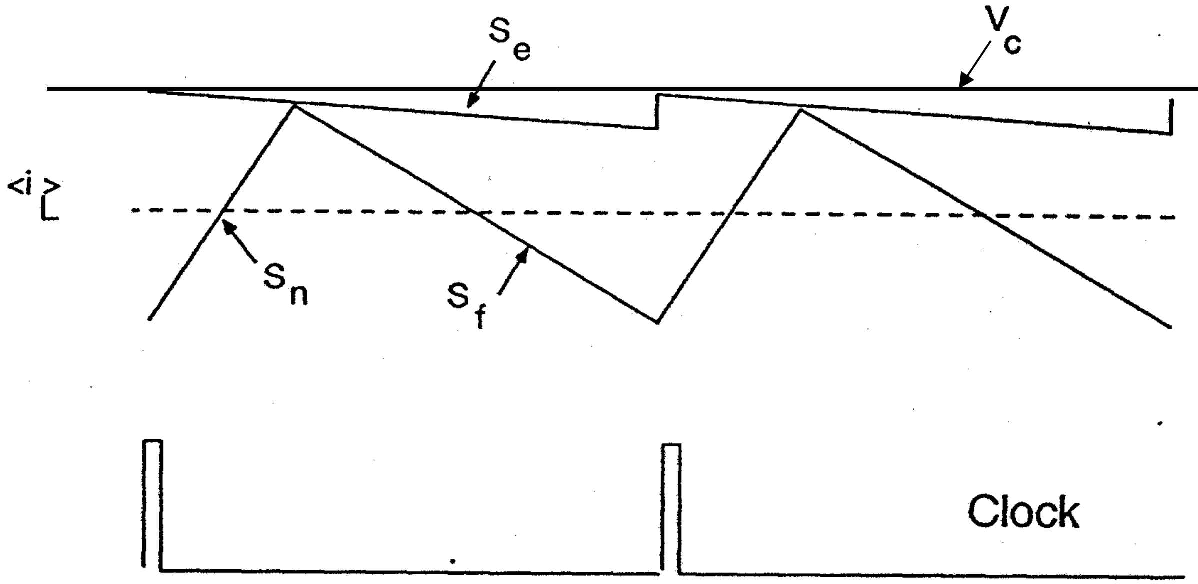
(a) Buck

- Voltage across the inductor is sensed, and integrated to provide a voltage proportional to the instantaneous inductor current.
- Reconstructed current signal is used as part of the modulator.

Modulator Waveforms for Current-Mode Control

Constant-Frequency Control, Clock Initiates On-Time

(Also referred to as constant-frequency with trailing-edge modulation)

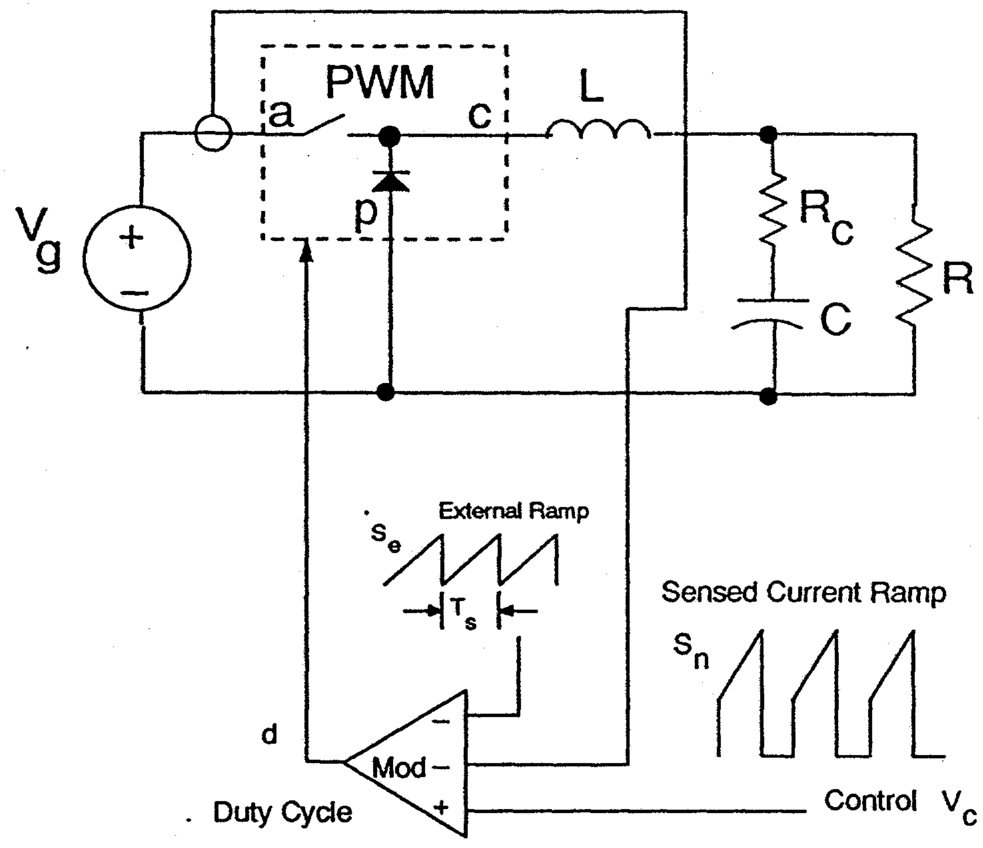


- **Clock signal triggers the turn on of the power switch.**
- **Switch turns off when the inductor current signal intersects the control voltage.**
- **Inductor current is used by the control**

Later implementation of current-mode control (1978)
"Current-Injection Control" (CIC) Implementation

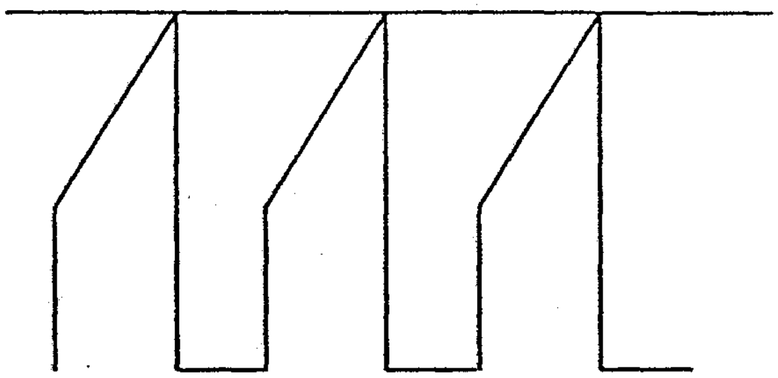
- Inductor current is sensed, usually with a current transformer or resistor in series with the power switch

- Diode current can be sensed for leading-edge modulation



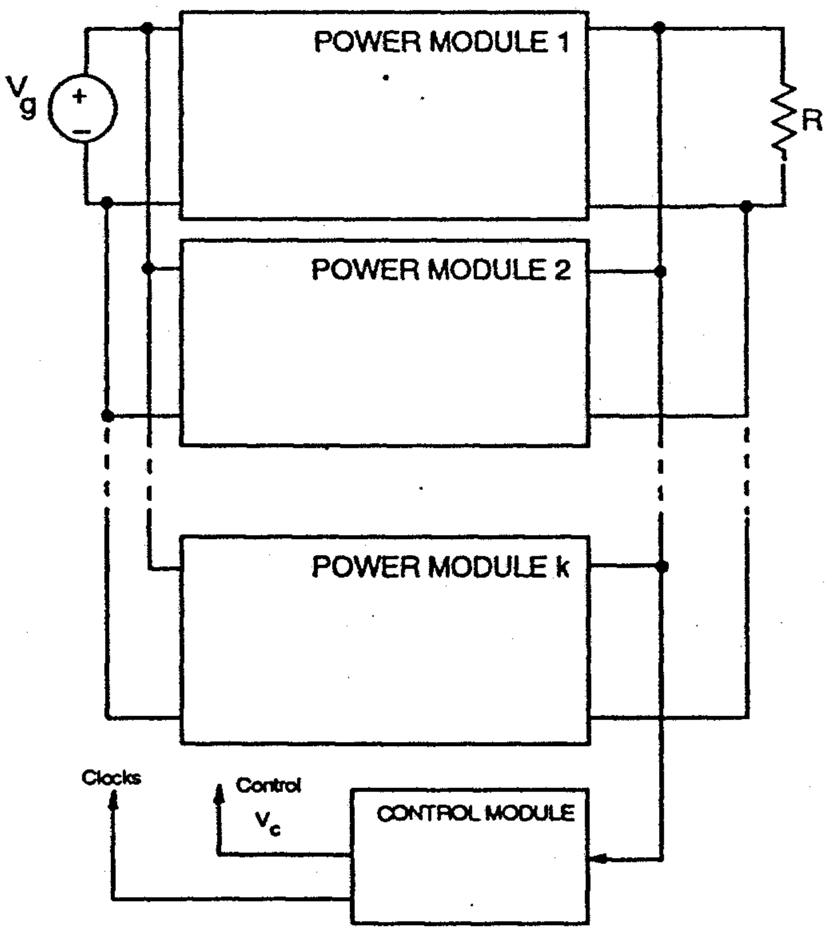
(a) Buck

Current limiting with CIC control



- Limited value of error voltage provides peak current protection
- No soft-start mechanism is required

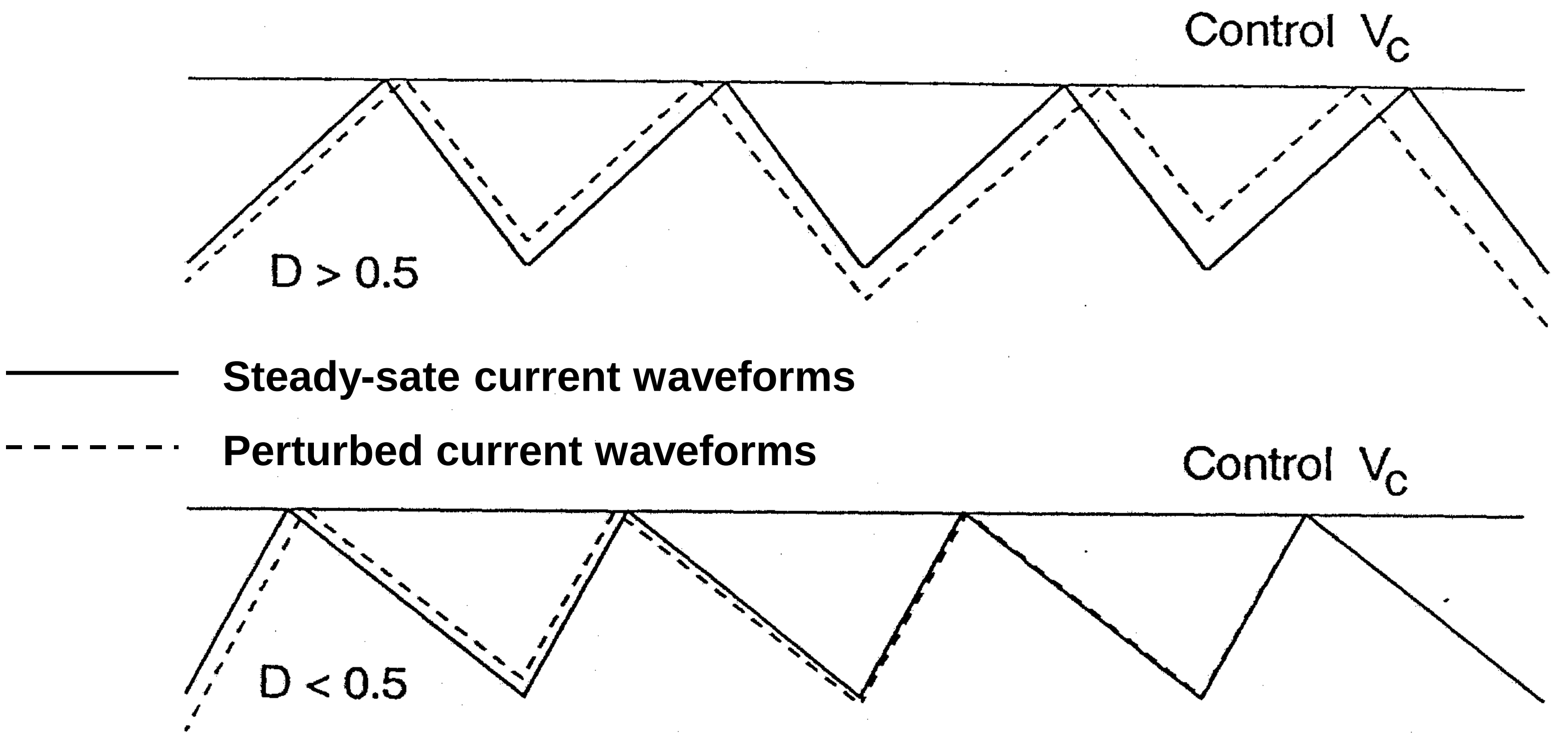
Current sharing with CIC control



- Modules share a common error voltage
- Peak current in each switch is kept the same

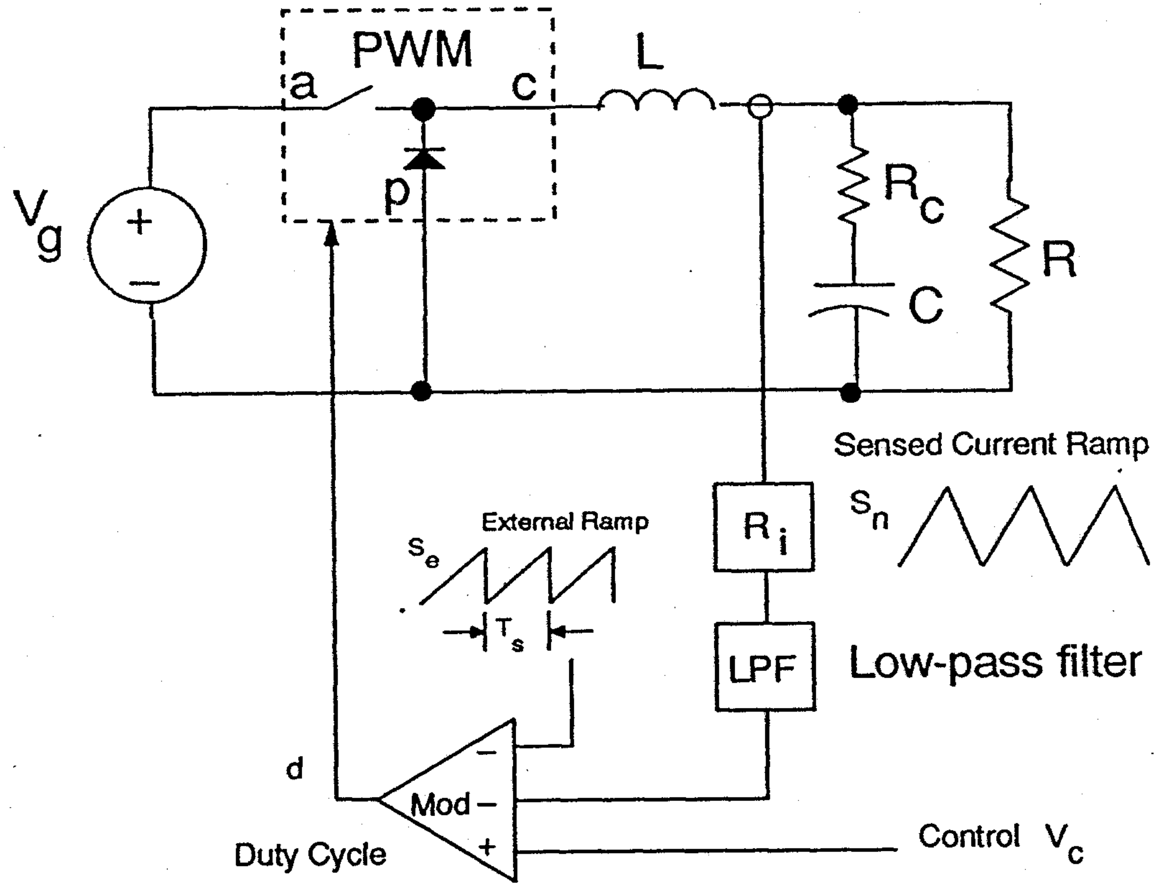
Instability Issue of Constant-Frequency Peak Current Mode Control

No External Ramp



- **Oscillation grows for duty cycle > 0.5**
- **Oscillation decays for duty cycle < 0.5**
- **Instability exists in current loop.**

Average current-mode control



- **Sensed inductor current is filtered to remove ripple component**

History of Current-Mode Control Modeling

- Averaging models

  Lee's modeling — Two pole system
  Middlebrook's modeling — Two-pole system (simplified to 1 pole)
  Unitrode's model — based on Middlebrook's 1-pole model
  MIT's model — based on a "correction" to Middlebrook's model
  Kislovski's "current-injection" model — One pole

NOTE: ALL existing average models fail to correctly predict the current-loop instability and closed-loop characteristics.

- Discrete-time models

  Exact models correctly predict stability
  Numerical methods require computers
  Difficult to design and measure systems

- Sampled-data models (A. Brown)

  Brown's model predicted instability for reduced-order buck, but was never completed or used. Numerical methods needed.

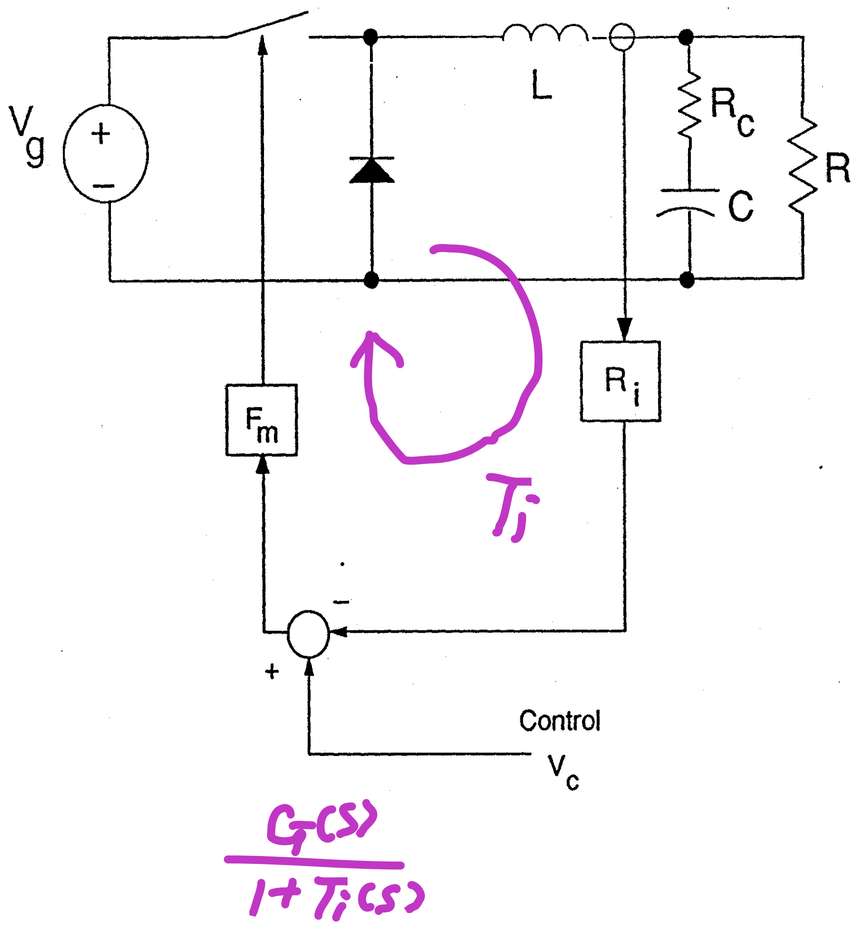
- Sampled-data model with the simplicity of pole-zero representation (R. Ridley)

  This model also predicted instability for reduced-order buck
  Accurate up to half the switching frequency
  Simple to use

Average Small-Signal Models

Buck Converter

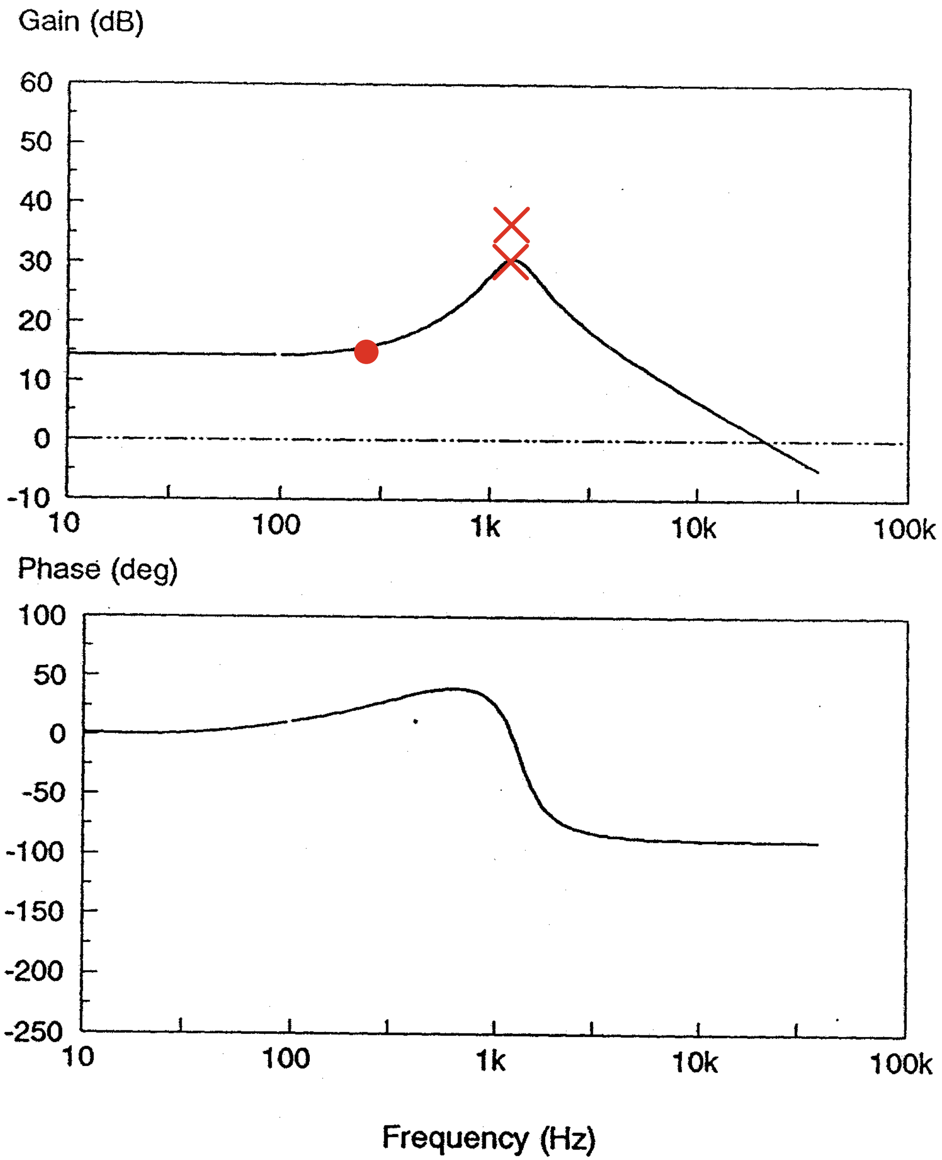
- Average current fed back with gain Ri
- "Inner" feedback loop is created with average model



Current-Loop Gain (Ti) with Average Model

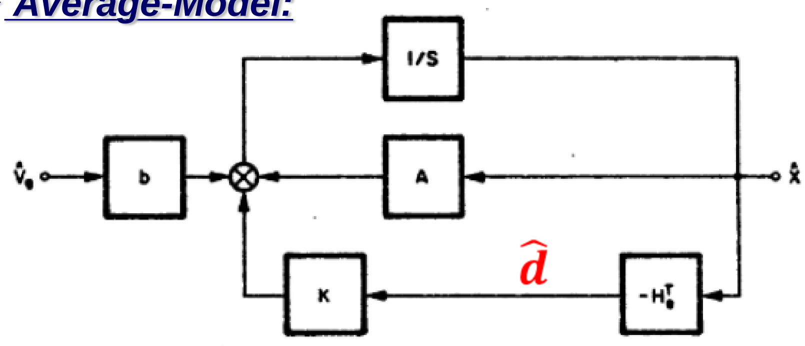
Worst-case phase delay is -90 degrees

No instability is predicted in current loop with average model



Sampled-Data Model Concept (Arthur Brown)

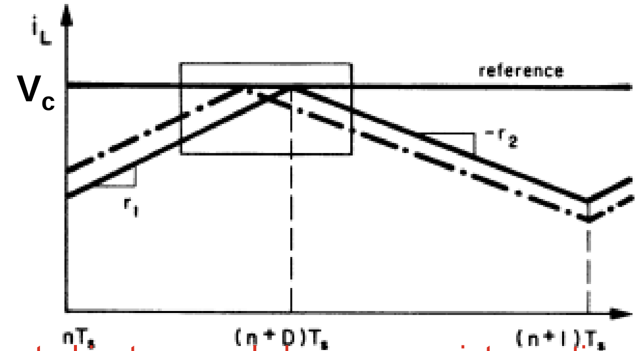
- **Average-Model:**



a) state-space averaging

Issue: Duty Ratio is Assumed to Continuously Change.

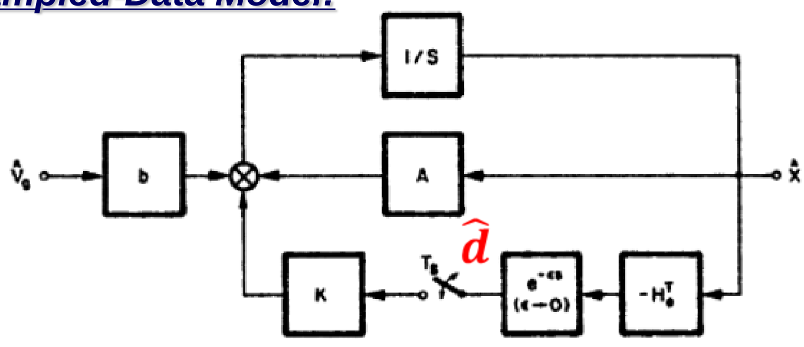
- **Modulator Waveforms:**



control instance only happens on intersection of vc to reference, performed as digital system

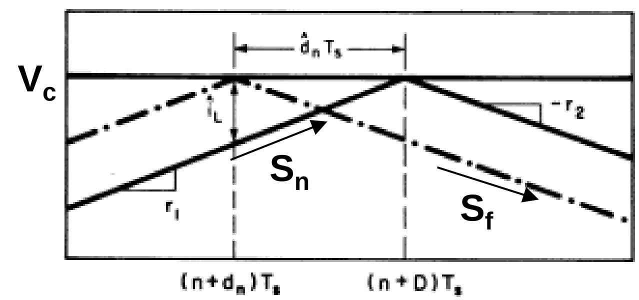
(a)

- **Sampled-Data Model:**



b) sampled-data modeling

- **Duty Ratio is Updated at One Single Instant in each switching cycle.**



(b) expanded view of (a)

steady-state
perturbed

Figure 7. *Current-programming modulator waveforms.*

Sampled-Data Approach Models PWM as a “Small-Signal” Sampler

A. R. Brown and R. D. Middlebrook, "Sampled-data modeling of switching regulators," *1981 IEEE Power Electronics Specialists Conference*, Boulder, CO, USA, 1981, pp. 349-369, doi: 10.1109/PESC.1981.7083659.

Sampled-Data Current-Loop Gain (T_i^*) (Arthur Brown)

➢ ***Sampled-Data Current Loop Gain:***

$$T_i^*(s) = \left(\frac{S_n + S_f}{S_n}\right)\left(\frac{1}{e^{sT_{sw}} - 1}\right) \quad (1)$$

➢ **S-Domain Model**

➢ **Sampled-Data Loop Gain Phase Reaches $-180°$ as Bandwidth Approach $f_{sw}/2$**

➢ **Can Predict instability By Showing a Negative Phase Margin**

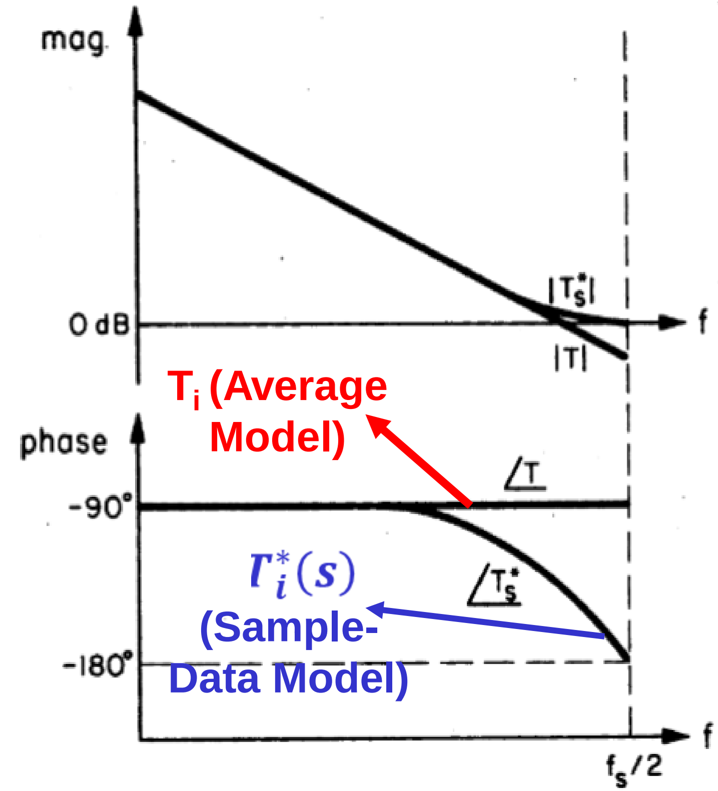


Figure 18. *State-space averaged and sampled-data loop gains for a single-pole system.*

A. R. Brown and R. D. Middlebrook, "Sampled-data modeling of switching regulators," *1981 IEEE Power Electronics Specialists Conference*, Boulder, CO, USA, 1981, pp. 349-369, doi: 10.1109/PESC.1981.7083659.

A. Brown's Explanation for Current-Loop Instability

➢ Closed-loop Pole Location:

$$\Rightarrow 1 + T_i^*(s_p) = 0 \qquad (2)$$

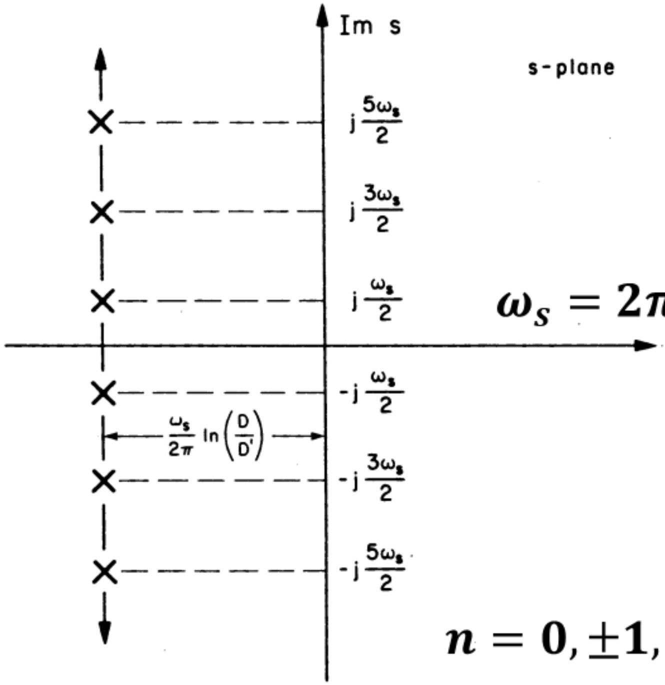
Subs. (1) in (2), we get

$$s_p = \underbrace{\frac{\omega_s}{2\pi}\ln\left(\frac{D}{D'}\right)}_{\mathrm{Re}(s_p)} + \underbrace{j\left(n+\frac{1}{2}\right)\omega_s}_{\mathrm{Im}(s_p)}$$

$$D > 0.5 \Rightarrow \frac{D}{D'} > 1 \Rightarrow Re(s_p) > 0$$

(Unstable)

➢ Pole Map: $D < 0.5$



$\omega_s = 2\pi/T_{sw}$

$n = 0, \pm 1, \pm 2, \ldots$

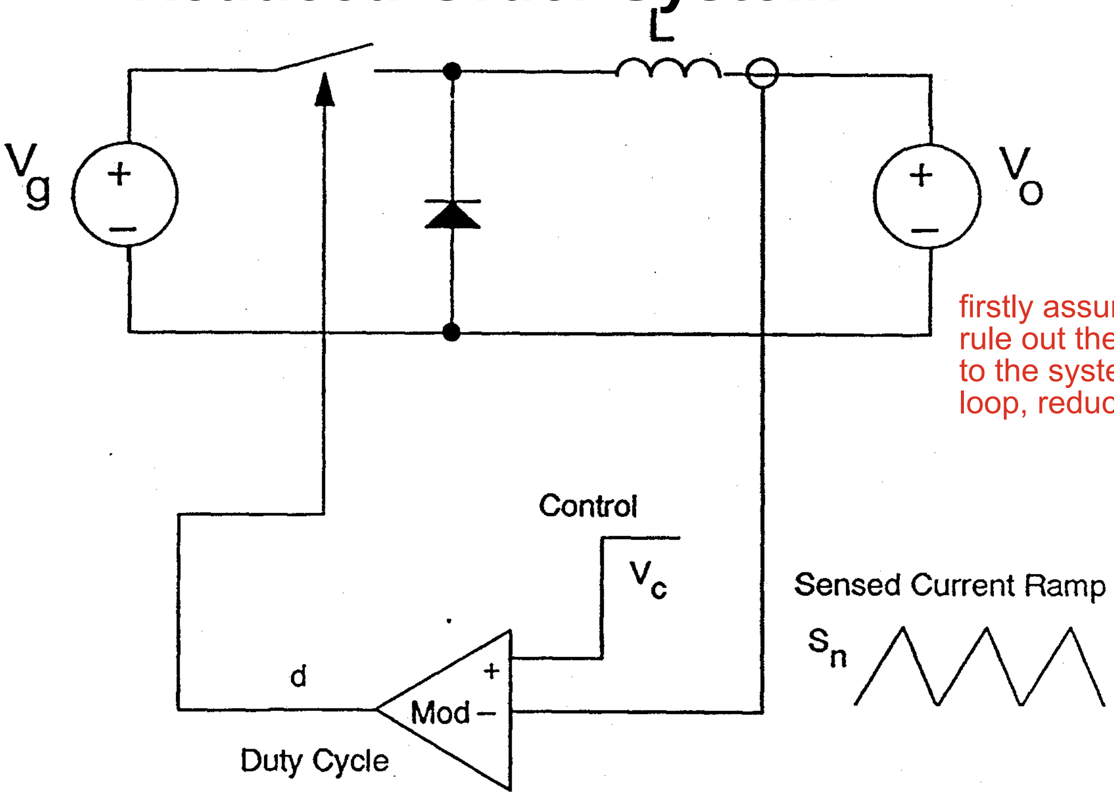
Figure 15. *Current-programming poles from a sampled-data analysis.*

Successfully Explained Instability for D>0.5 in "s-Domain"

A. R. Brown and R. D. Middlebrook, "Sampled-data modeling of switching regulators," *1981 IEEE Power Electronics Specialists Conference*, Boulder, CO, USA, 1981, pp. 349-369, doi: 10.1109/PESC.1981.7083659.

Discrete-Time Analysis

Reduced-Order System



firstly assum vo is voltage source rule out the output voltage factor to the system, we only study current loop, reduced system order

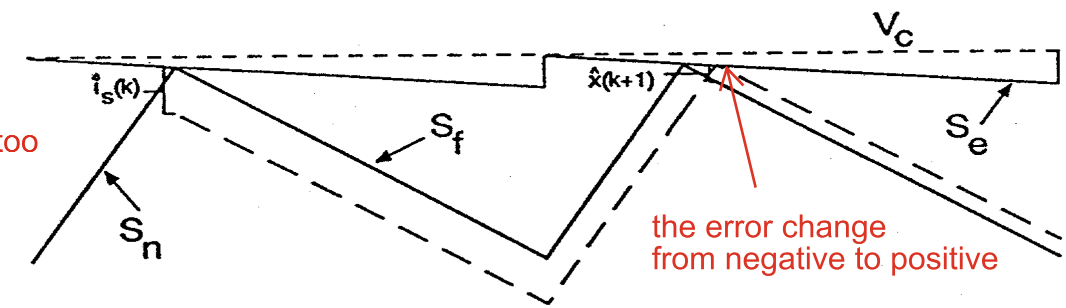
- Reduced order for analytical results
- Instability clearly show

Raymond B. Ridley "A new small-signal model for current-mode control," Ph.D. dissertation, Virginia Polytechnic Institute and State University, Blacksburg, Nov. 1990.

Natural Response
of Discrete-Time System

When Se is too large compared to inductor ripple, the system behavior will fall into voltage mode:

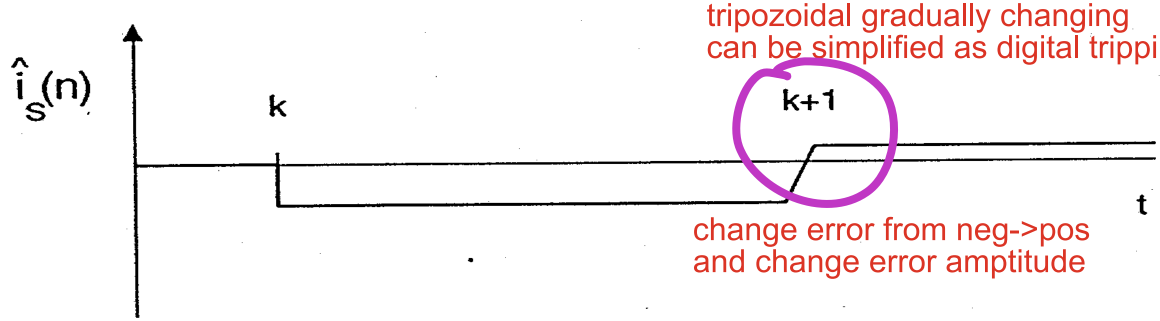
the current loop controllability is weak (iL ripple is too small compared to Se slope), the inductor current loop will relinquish and only vc control the system which is back to voltage mode control



the error change from negative to positive

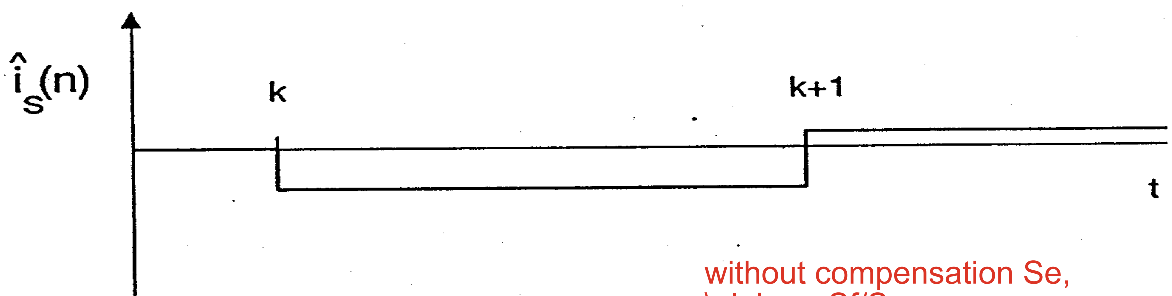
tripozoidal gradually changing can be simplified as digital tripping

Exact response of current-signal



change error from neg->pos and change error amptitude

Approximate response of current-signal



natural responds of the system

Homogeneous response:

$$\hat{i}_L(k+1) = -\alpha \hat{i}_L(k)$$

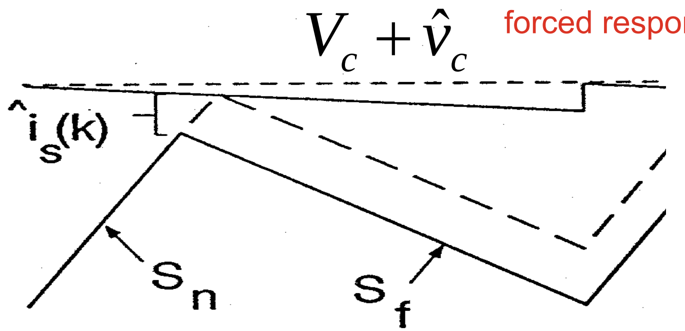
without compensation Se, \alpha = Sf/Sn

$$\boxed{\alpha = \frac{S_f - S_e}{S_n + S_e}}$$

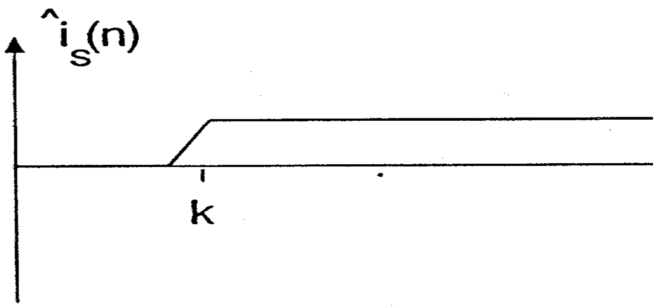
Forced Response

of Discrete-Time System

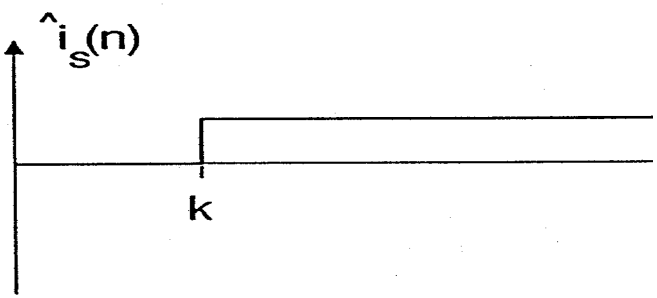
forced responds of system (vc change)



Exact response of current-signal



Approximate response of current-signal



Forced response:

$$\hat{i}_L(k) = \frac{1}{R_i}(1+\alpha)\hat{v}_c(k) \quad \Rightarrow \quad \hat{i}_L(k+1) = \frac{1}{R_i}(1+\alpha)\hat{v}_c(k+1)$$

Z-Transform
and Discrete-Time Poles

Complete response:

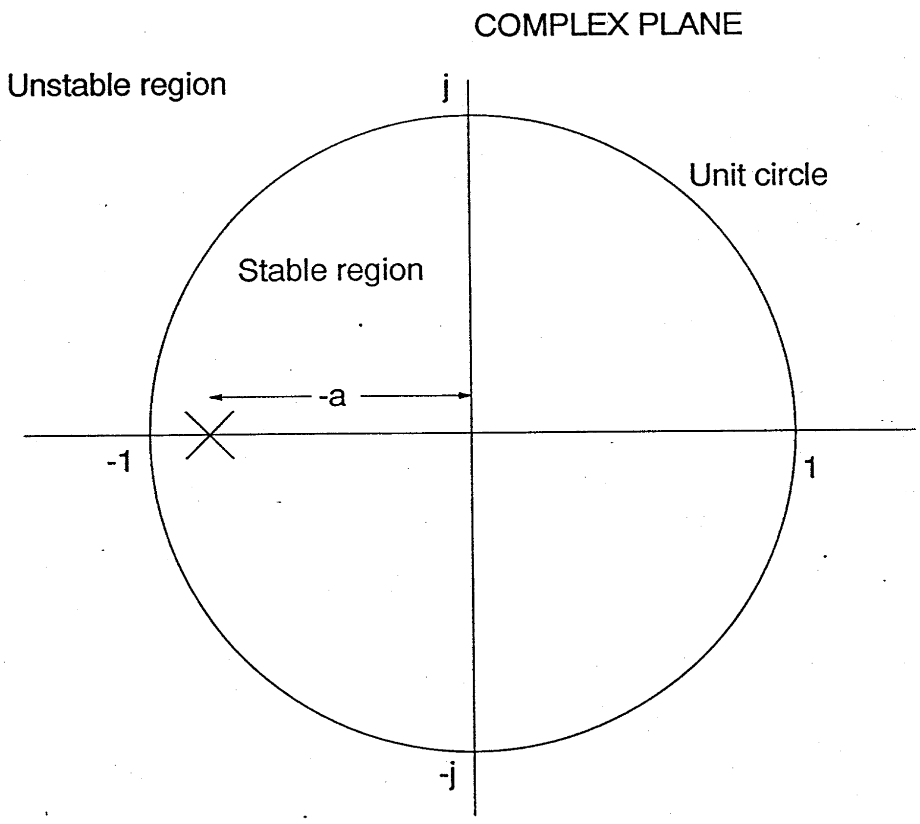
$$\hat{i}_L(k+1) = -\alpha\,\hat{i}_L(k) + \frac{1}{R_i}(1+\alpha)\hat{v}_c(k+1)$$

natural responds   forced responds

Z-transform:

$$\boxed{H(z) = \frac{\hat{i}_L(z)}{\hat{v}_c(z)} = \frac{1}{R_i}(1+\alpha)\frac{z}{z+\alpha}}$$

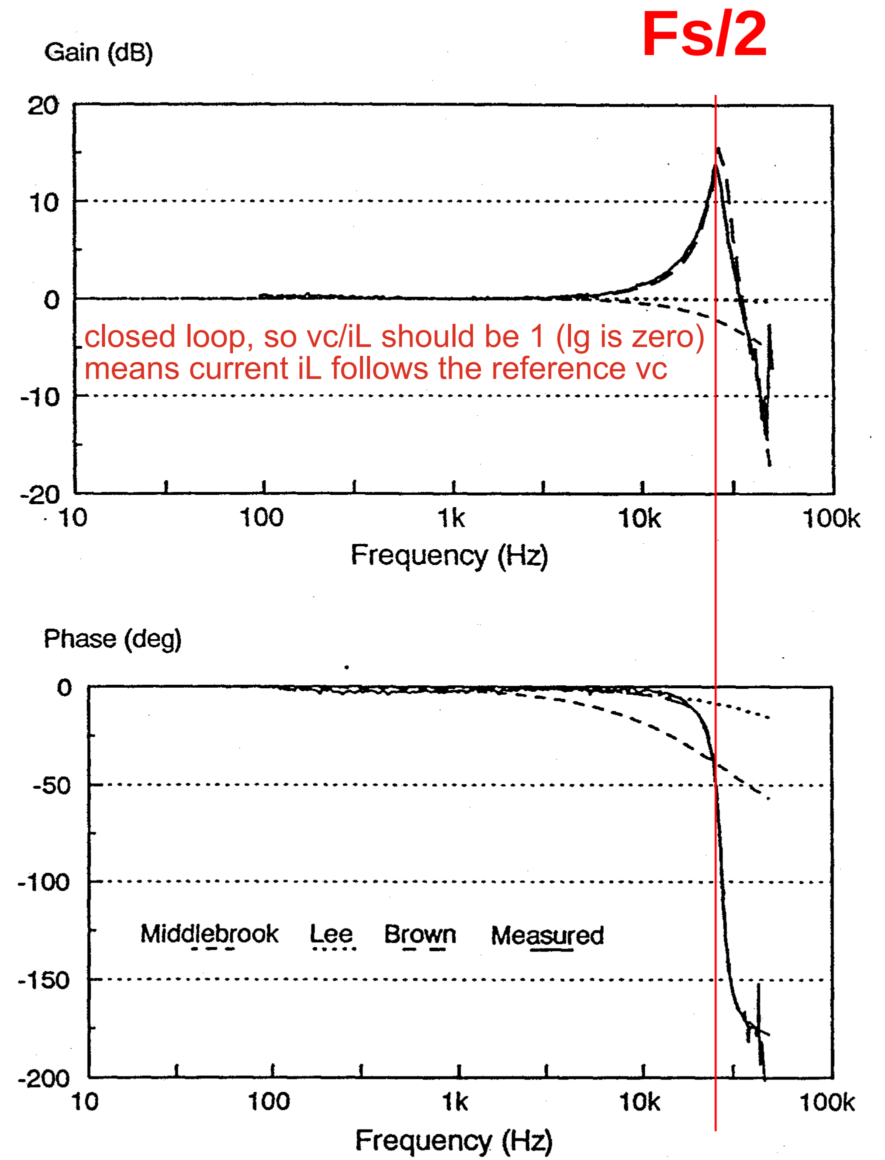
closed loop transfer fuction vc-> iL equation
the control system already closed its loop



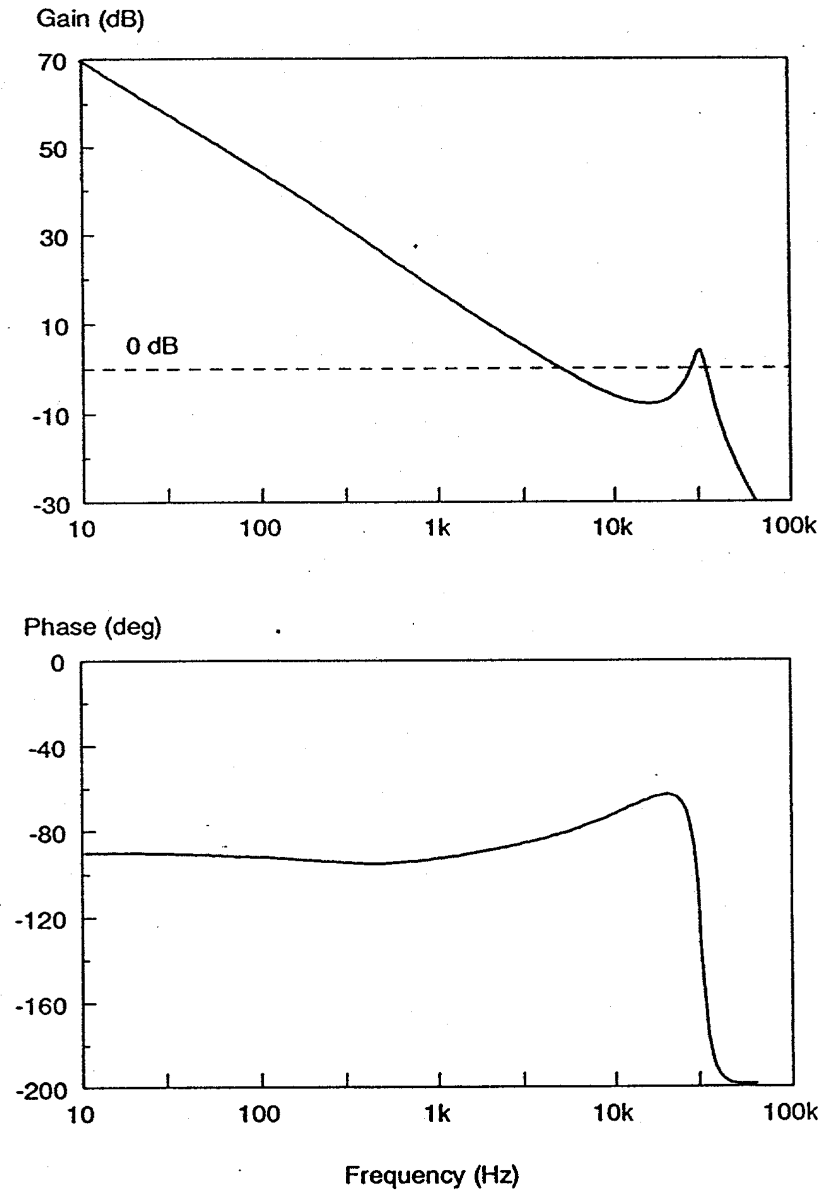
- **Discrete-time analysis clearly shows instability**
- **Only single-pole system has been modeled**

Close-loop v_c to i_L (control to inductor current)
Buck Converter, D=0.45

- Average models work well up to half the switching frequency
- Average models have large deviations at half switching frequency
- Sampled data model is accurate beyond half switching frequency
- Sampled data model is abandoned due to complexity



Voltage Loop Gain (T_2) with Current Loop Closed (Buck Converter)



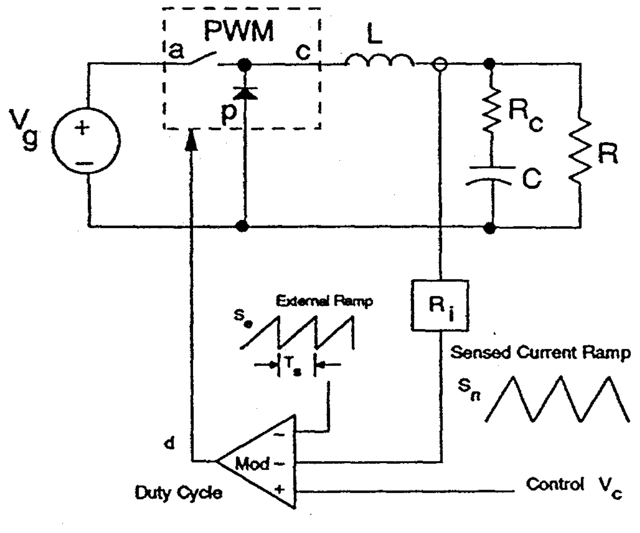
There is a peaking at Fs/2, which may cause the second cross-over

- **High-frequency model is required even when low-bandwidth system is designed**

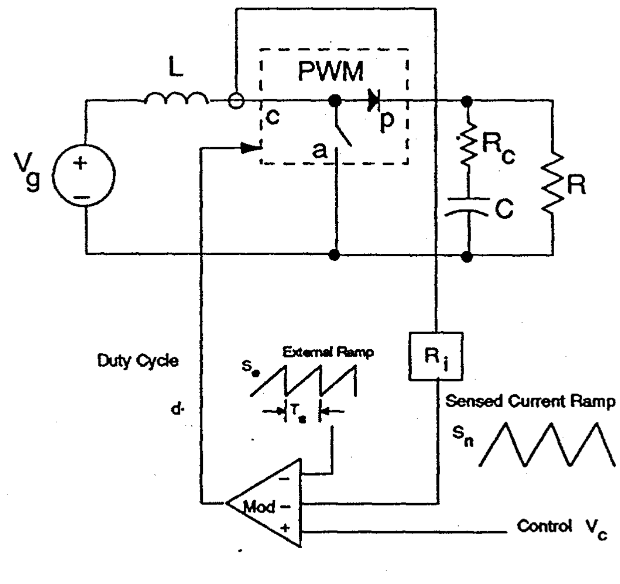
Sampled-data model with the simplicity of pole-zero representation (R. Ridley)

Power Stage Modeling

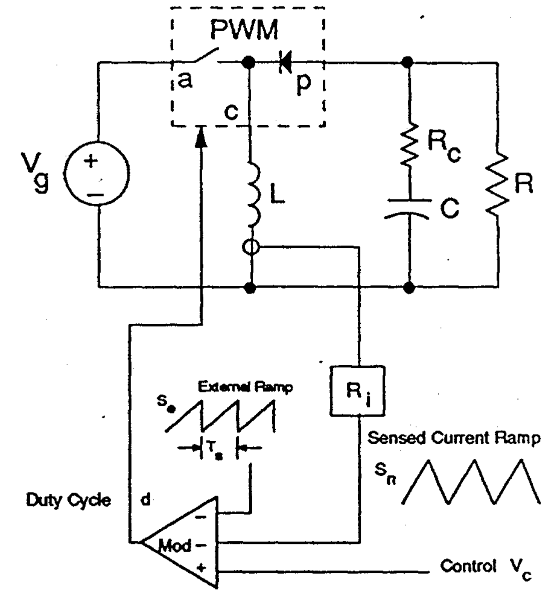
Step 1



(a) Buck



(b) Boost

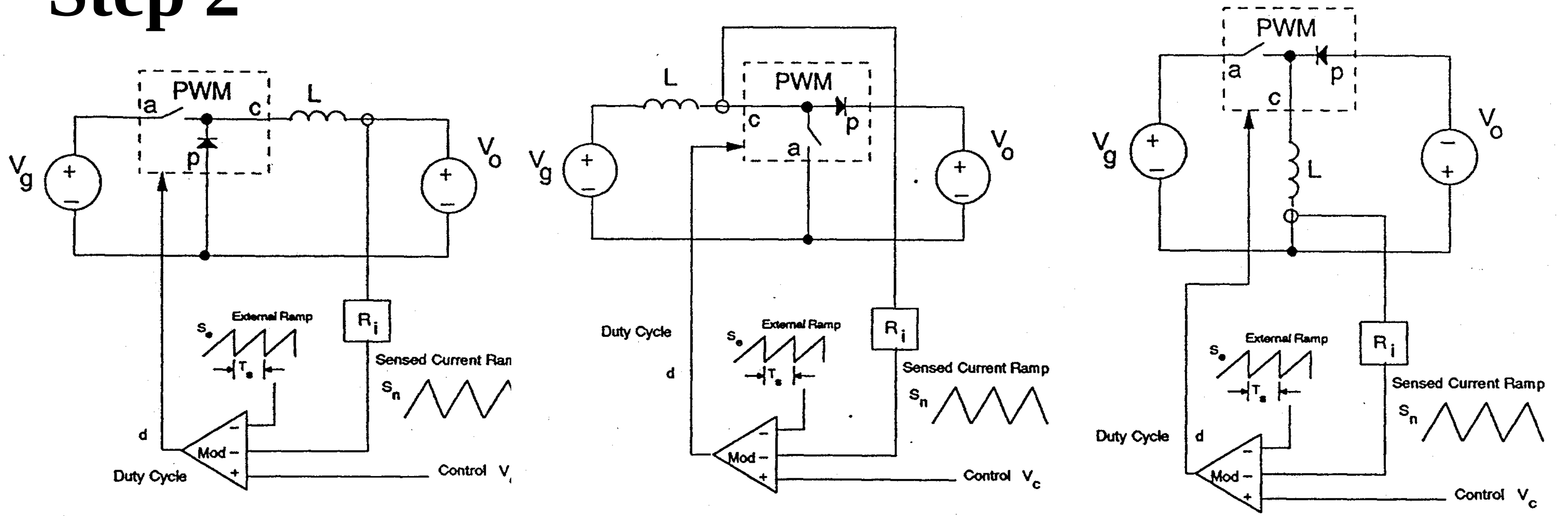


(c) Buck-Boost

- Power stage is unchanged by the presence of feedback loop
- Same small-signal model should be used for the power stage as with voltage-mode control
- PWM switch model approach is used for small-signal of power stage

PWM Current-Mode Switch Cell

Step 2

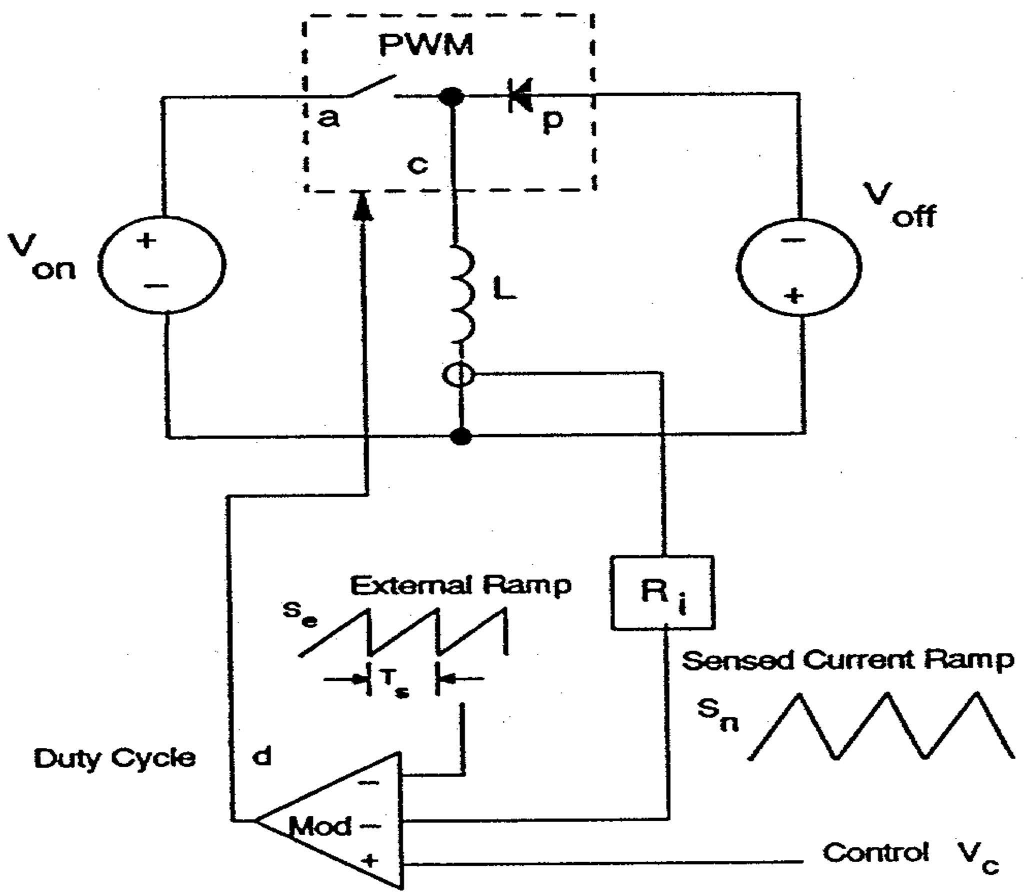


(a) Buck (b) Boost (c) Buck-Boost

- **High-frequency model of the ‘current cell’ only is needed**
- **Invariant structure can be identified with fixed voltages around the current cell**

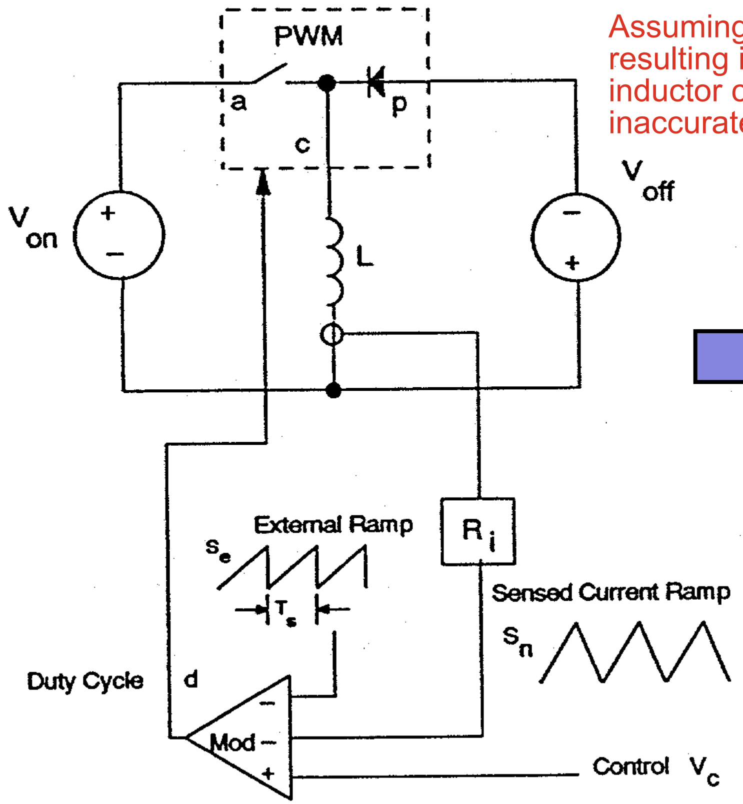
Generic Current Cell

Step 3

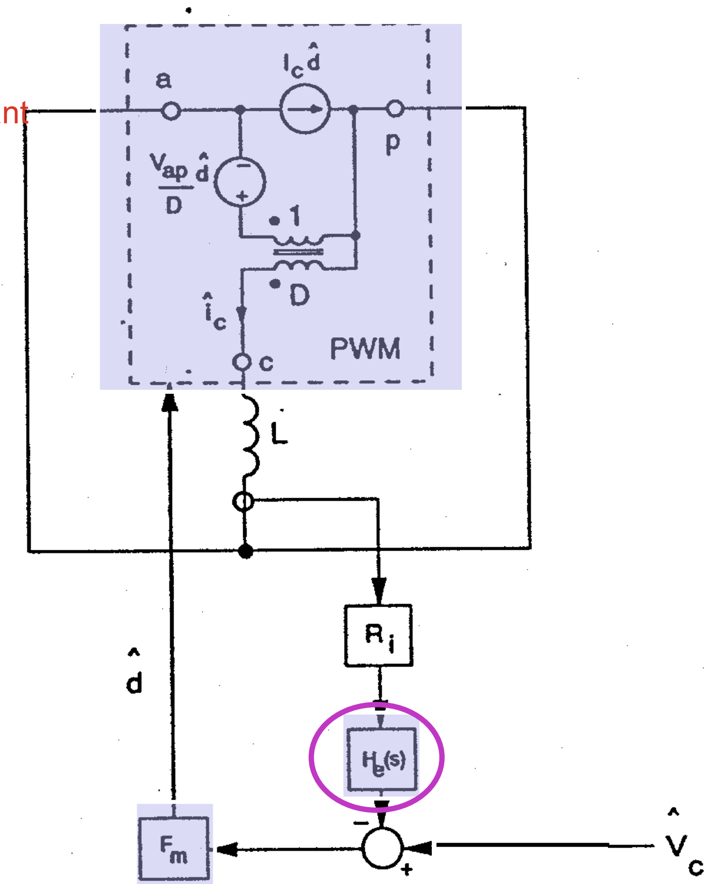


Power Stage Model for Generic Current Cell

Step 4



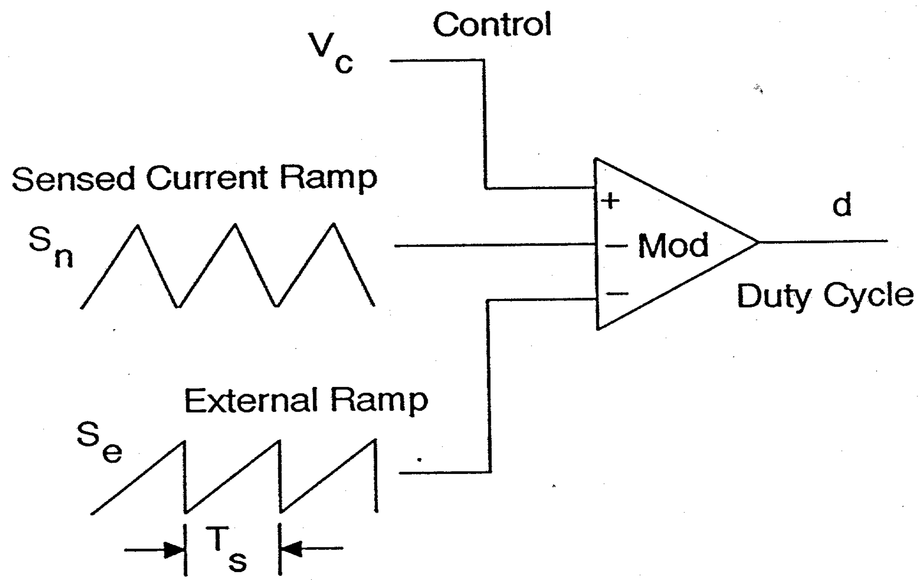
Assuming the Vo is constant resulting in fixed slope on inductor current, which is inaccurate



Assumption 1: 3-Terminal Switch model is still valid

Pulse-Width Modulator Model

Step 5:

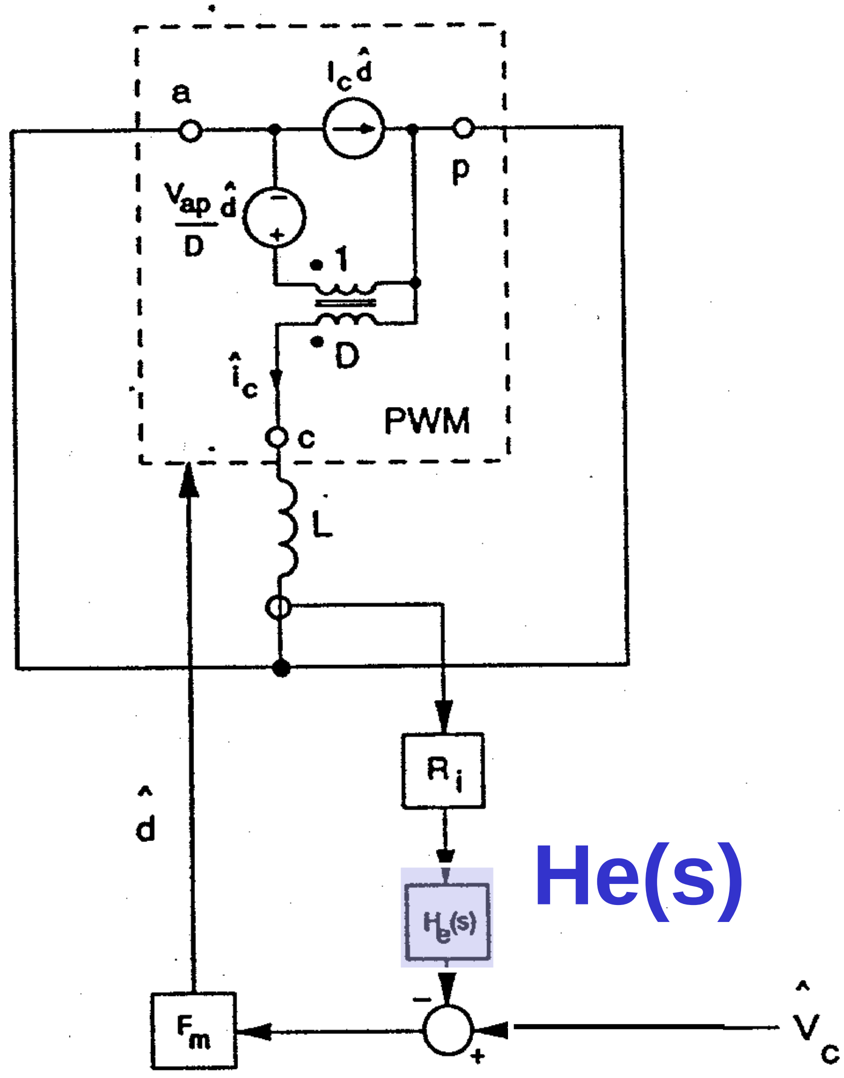


$$F_m = \frac{1}{m_c S_n T_s} \qquad m_c = 1 + \frac{S_e}{S_n}$$

Assumption 2: The PWM gain can be computed the same way as voltage–mode control

Account for the Sample-Data Effect by He(s)

Step 6

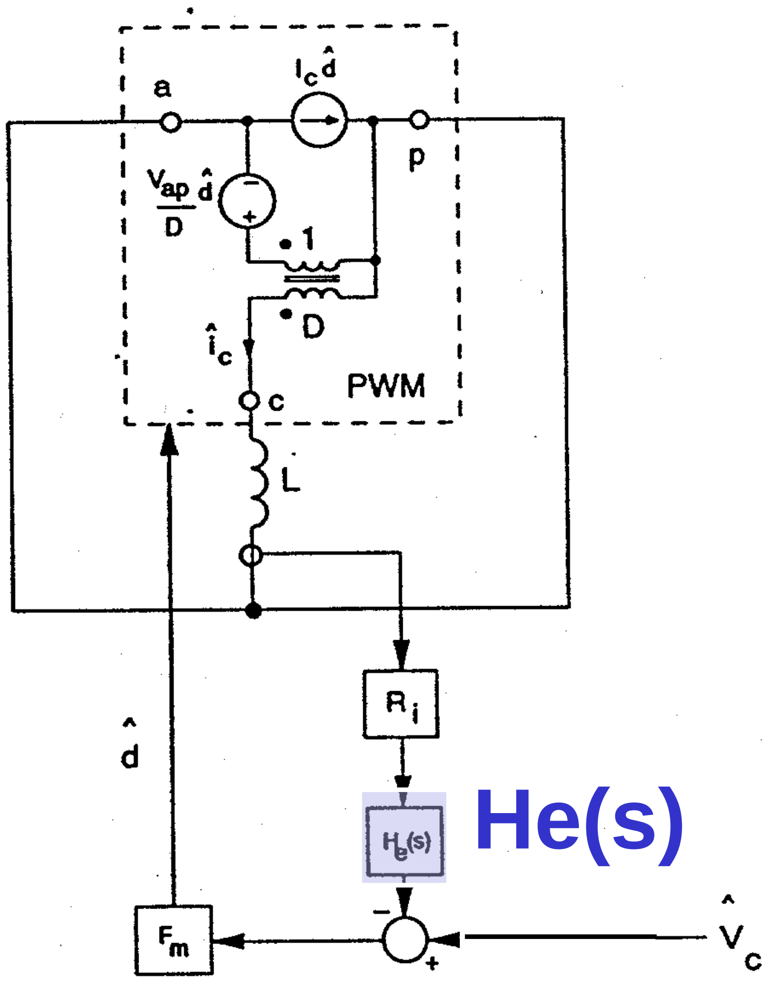


Assumption 3: He(s) is after R_i

He(s) represents sample-Data effect (Sample & hold effect)

Account for The Sample-Data Effect by He(s)

Step 6



From sample-data model (Brown's results)

$$\frac{\hat{i}_L(z)}{\hat{v}_c(z)} = \frac{1}{R_i}(1+\alpha)\frac{z}{z+\alpha}$$

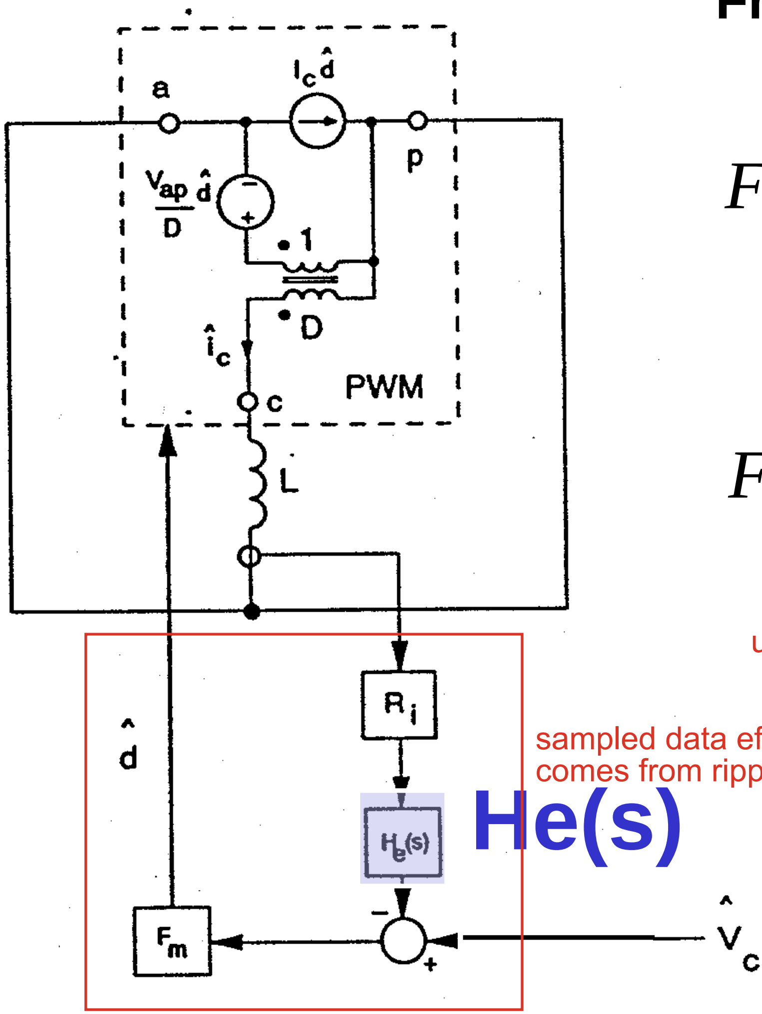
Transformation by substituting for z, and multiplying by sample and hold function

Continuous-time small-signal model

$$F(s) = \frac{\hat{i}_L(s)}{\hat{v}_C(s)}$$

$$= \frac{1}{R_i}\frac{1+\alpha}{sT_s}\frac{e^{sT_s}-1}{e^{sT_s}+\alpha} \qquad \textbf{(1)}$$

Open-Loop Sampling Gain He(s)



sampled data effect
comes from ripple

From the circuit, this is also given by:

$$F(s)=\frac{\hat{i}_L(s)}{\hat{v}_C(s)}=\frac{F_m F_i(s)}{1+F_m F_i(s) R_i H_e(s)} \qquad (2)$$

open loop TF

closed loop TF

$$F_i(s)=\frac{\hat{i}_L(s)}{\hat{d}(s)}$$

use z domin result in (1) to back induce the He(s) expression

Equating (1) and (2), we can solve for $H_e(s)$

$$\mathbf{H_e(s)=\frac{sT_s}{e^{sT_s}-1}}$$

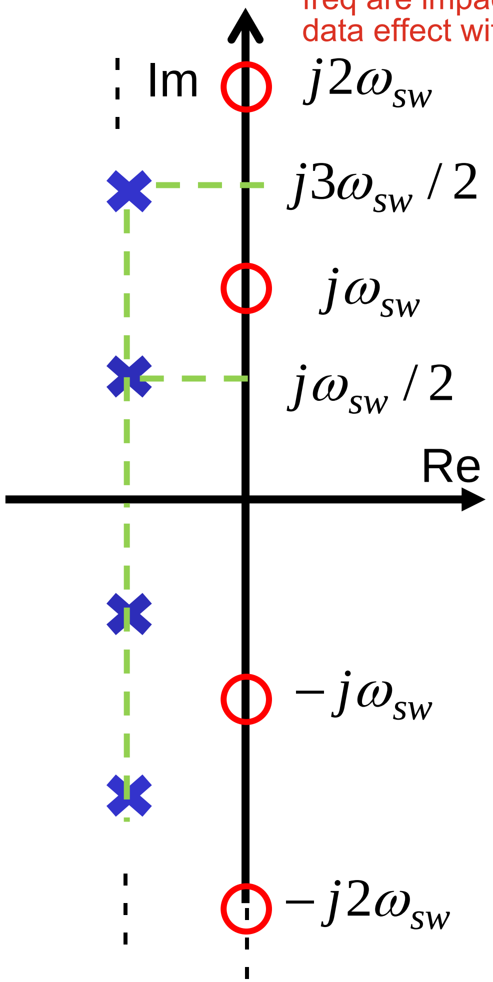
this part work as a whole, so He(s) can
be everywhere in this block, but the result
of block must be the same in both Eq (1) and (2)

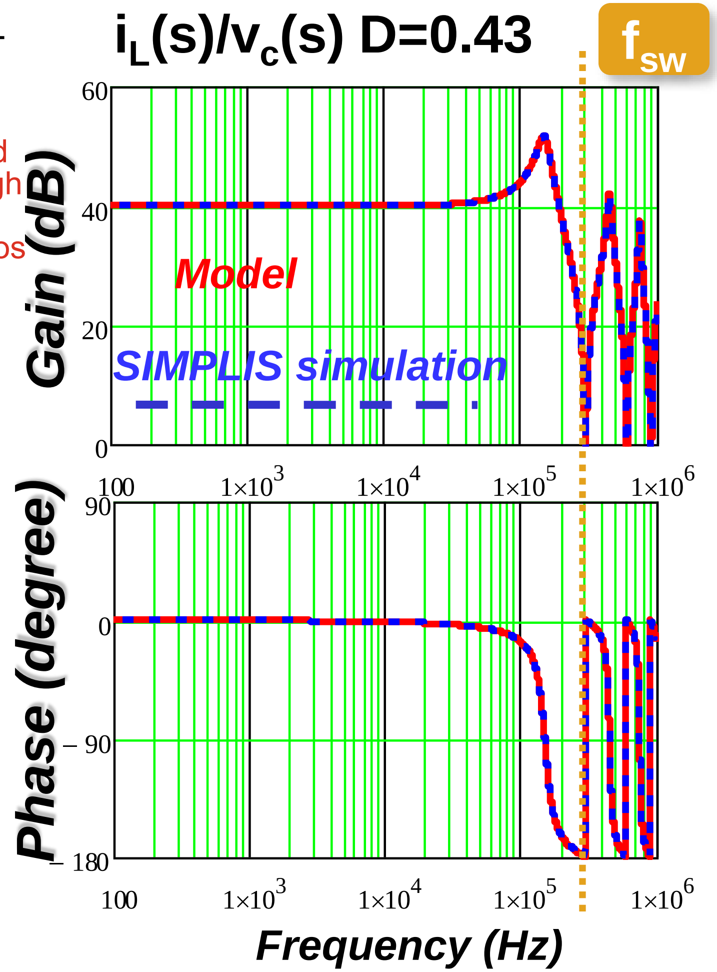
Control to Inductor Current

$$\frac{\hat{i}_L(s)}{\hat{v}_C(s)} = \frac{1}{R_i}\,\frac{1+\alpha}{sT_s}\,\frac{e^{sT_s}-1}{e^{sT_s}+\alpha}$$

flat gain at low freq, performed like a good current source, high freq are impacted by sampled data effect with poles and zeros

S-Plane





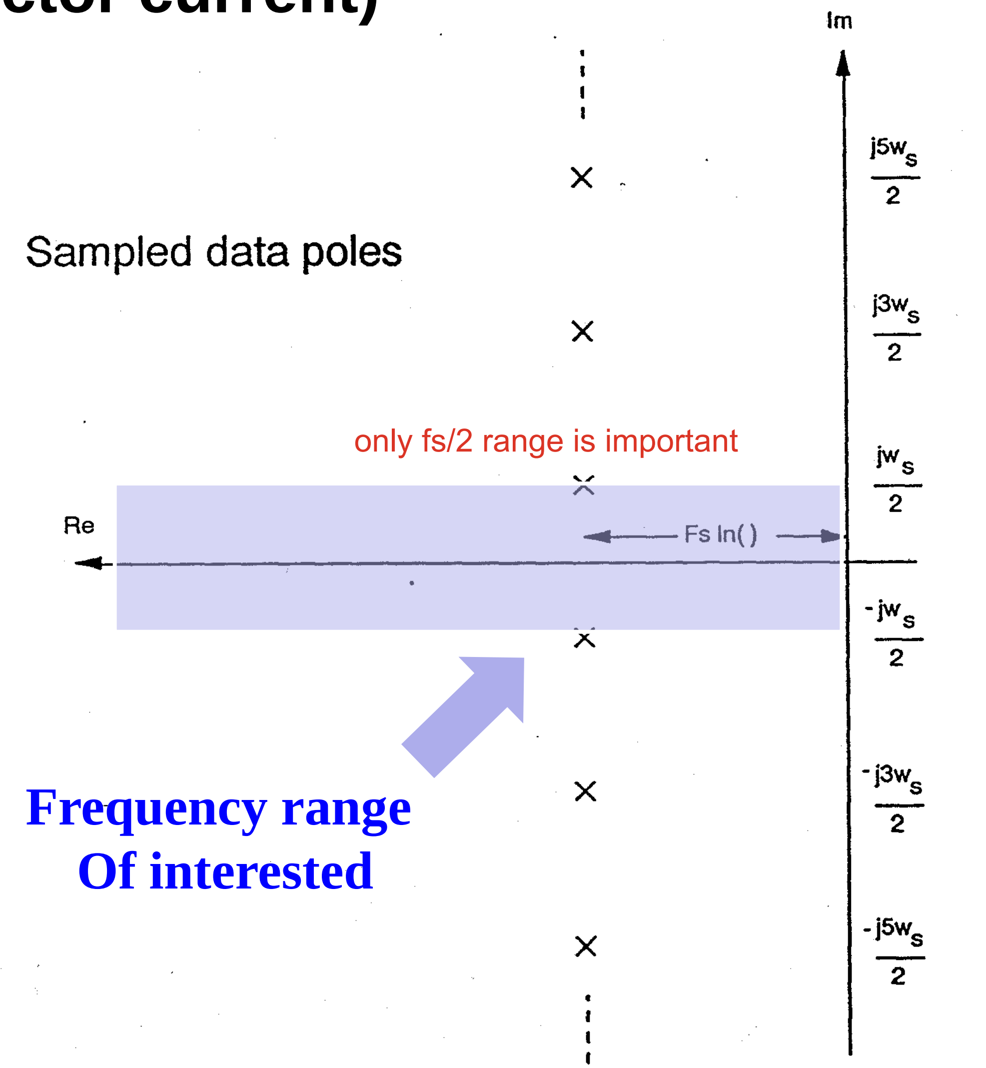
Dip is caused by zeros

peaks are caused by poles

Poles of Sampled-Data Expression

(Control-to-inductor current)

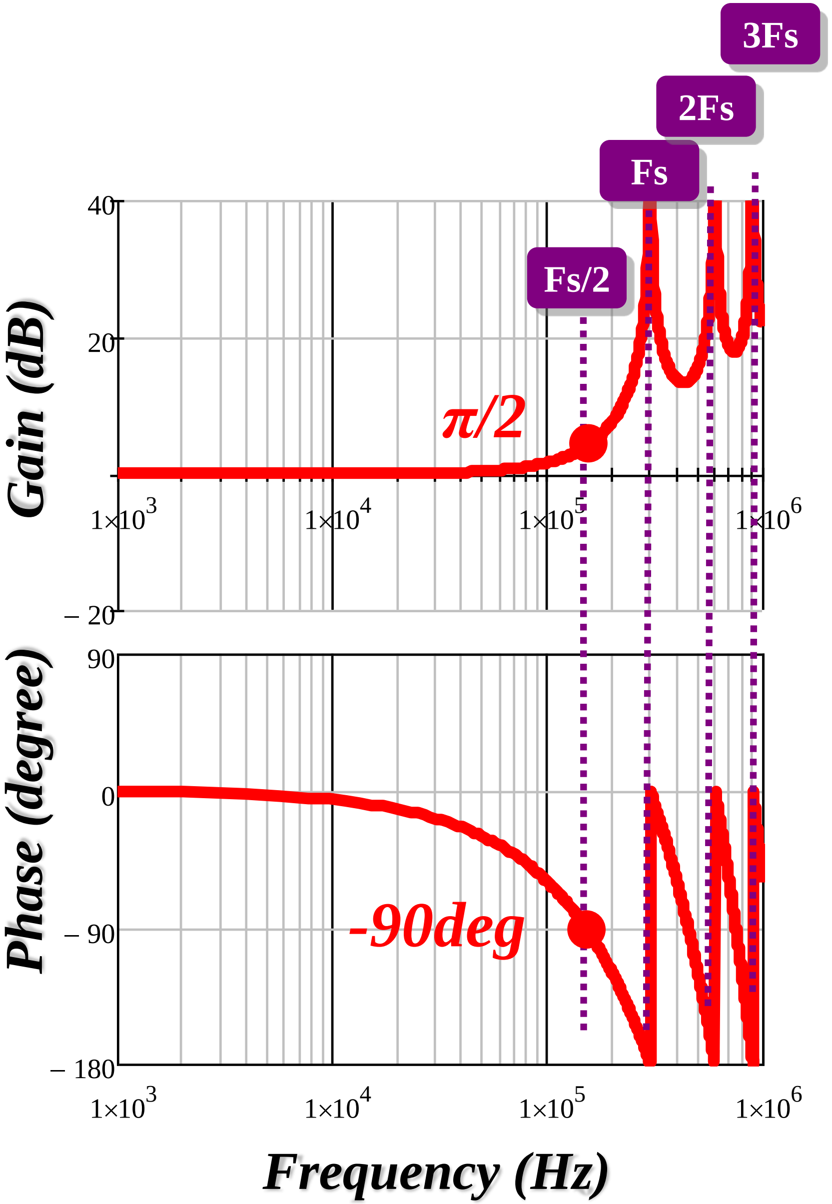
- **Infinite number of poles**
- **For all practical concern, only the lowest set of poles are important**



Pole-Zero Location and Bode Plot of $H_e(s)$

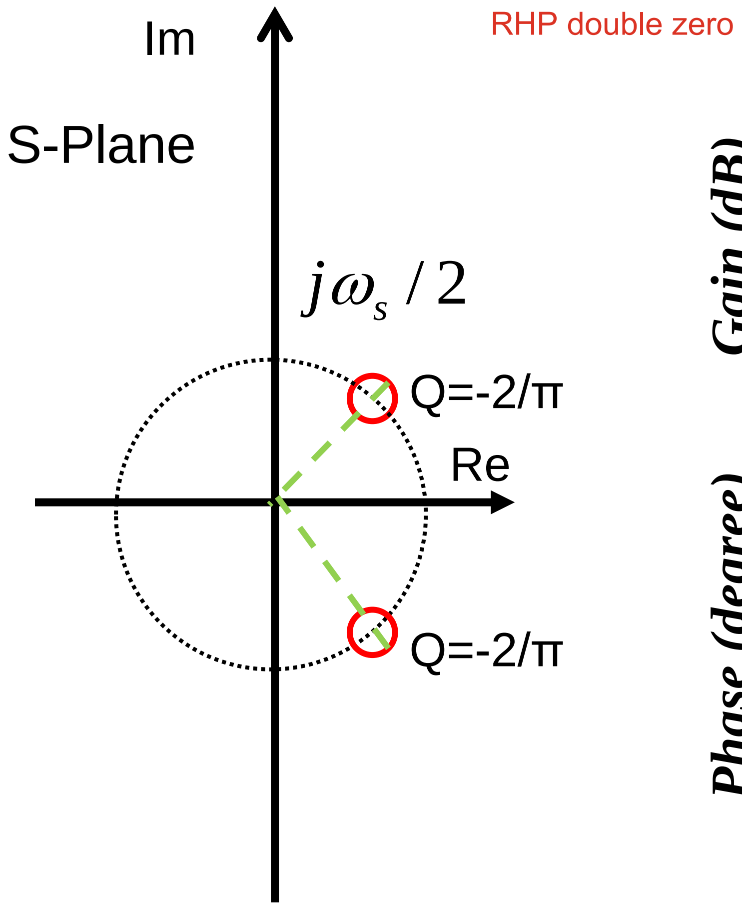
$$H_e(s) = \frac{sT_s}{e^{sT_s} - 1}$$

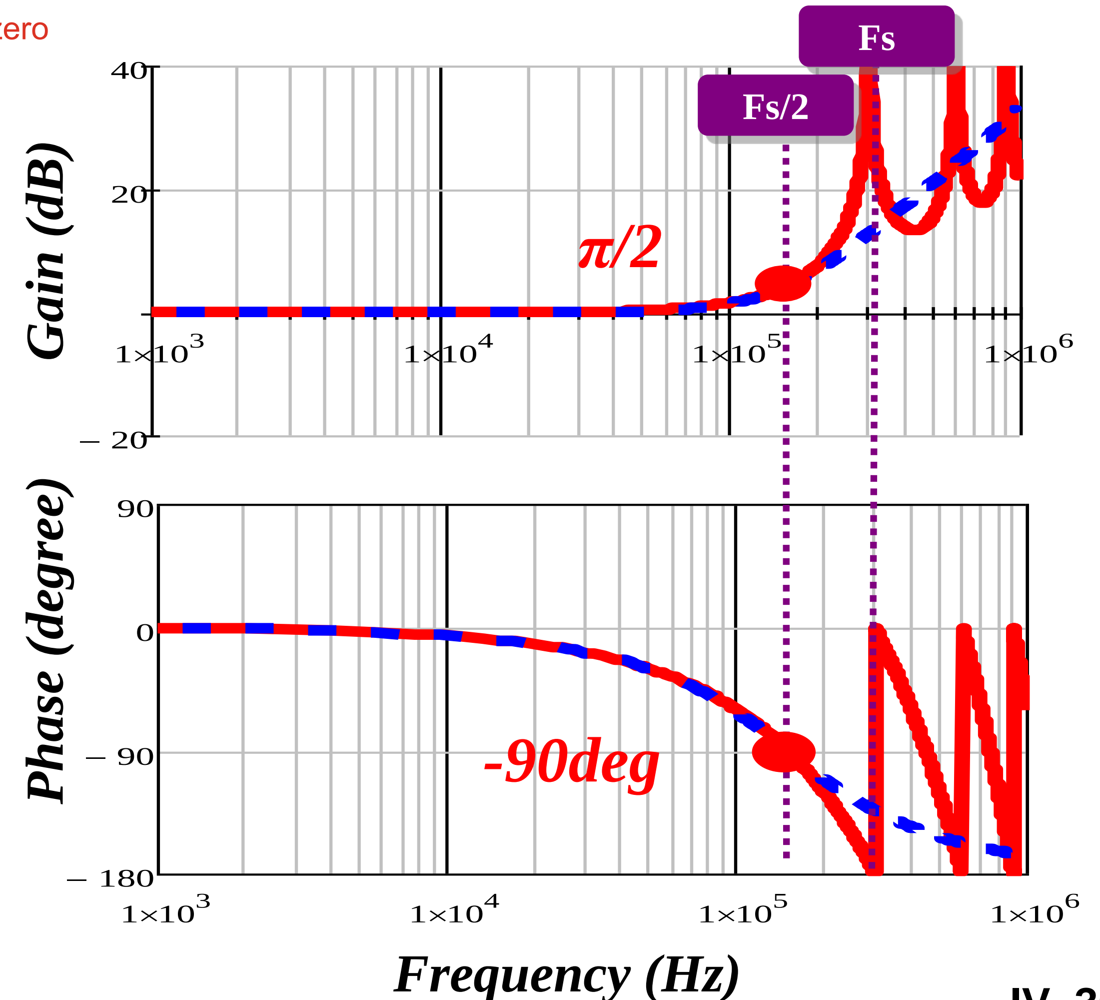
e term will eliminate the zero at dc freq, resulting flat gain and 0 phase



Pole-Zero Locations and Bode Plot of $H_e(s)$

$$H_e(s) \cong 1 + \frac{s}{\omega_n Q_z} + \left(\frac{s}{\omega_n}\right)^2 \qquad Q_z = -\frac{2}{\pi}, \omega_n = \frac{\pi}{T_s}$$





The Sampling Gain He(s)
-- A Quadratic Approximation

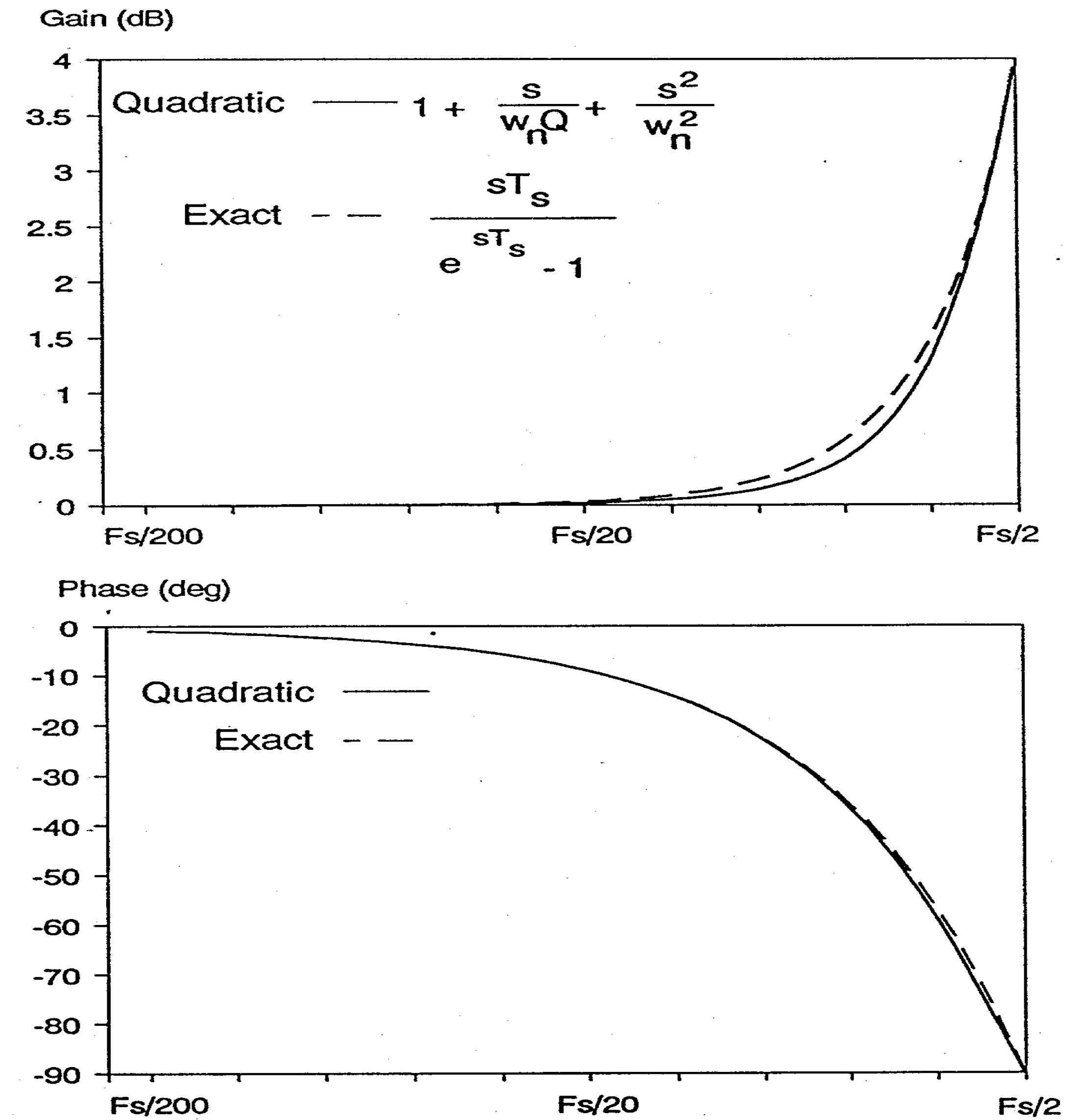
Gain and phase match exactly at DC and half the switching frequency.

$$H_e(s) = \frac{sT_s}{e^{sT_s} - 1}$$

$$\approx 1 + \frac{s}{w_n Q_z} + \frac{s^2}{w_n^2}$$

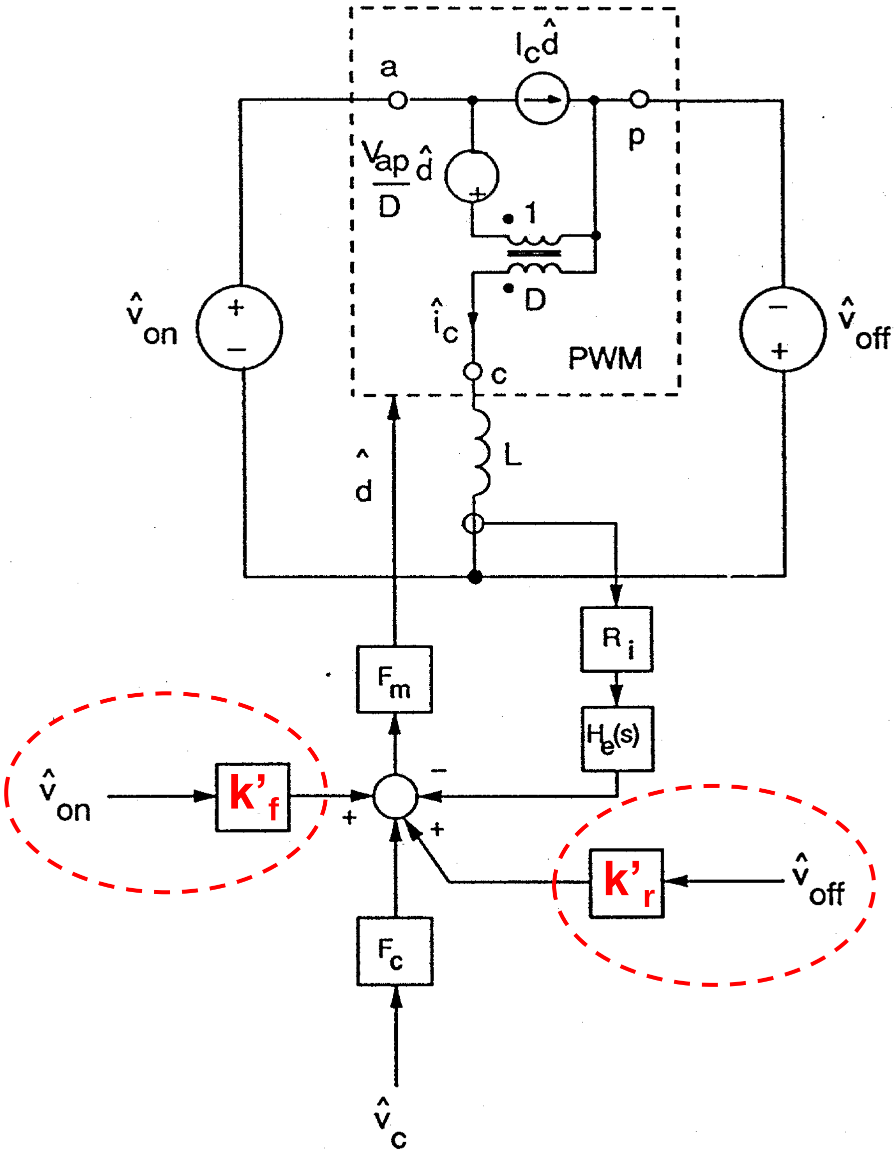
$$\omega_n = \frac{\pi}{T_s}$$

$$Q_z = \frac{-2}{\pi}$$



Complete Small-Signal Model for Peak-Current-Mode Control (R. Ridley)

- Perturbations in input and output voltages are introduced
- Gain blocks k'_f and k'_r are selected to provide correct response to voltage perturbations



Derivation of
Feedforward Gain K_f' and K_r'

— WHY?

V_{on} and V_{off} affect S_n and S_f respectively. These effects need to be taken into consideration.

— OBSERVATION:

In instantaneous current-mode control, inductor peak current is controlled. S_n and S_f do NOT affect peak current, but average current.

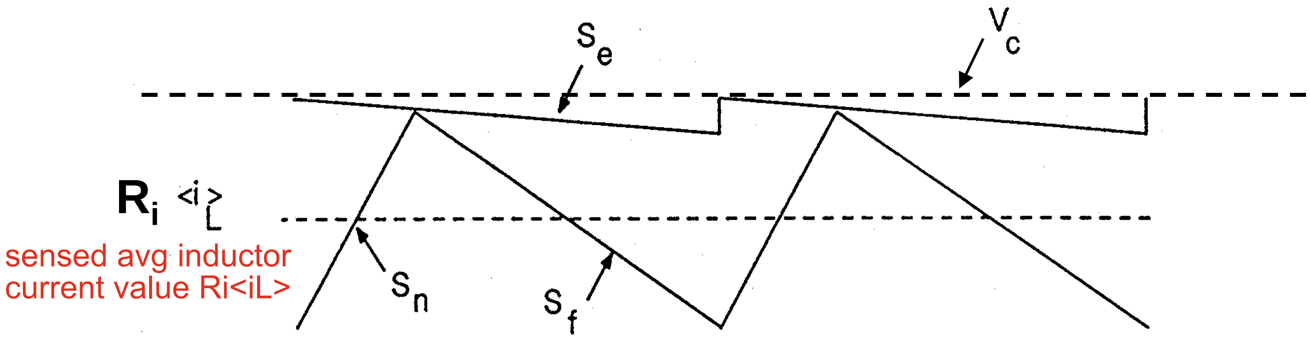
— DERIVATION METHOD:

From circuit waveform $\Rightarrow$ Equation 1

From block diagram $\Rightarrow$ Equation 2

Equ. 1 = Equ. 2 $\Rightarrow$ K_f' and K_r'

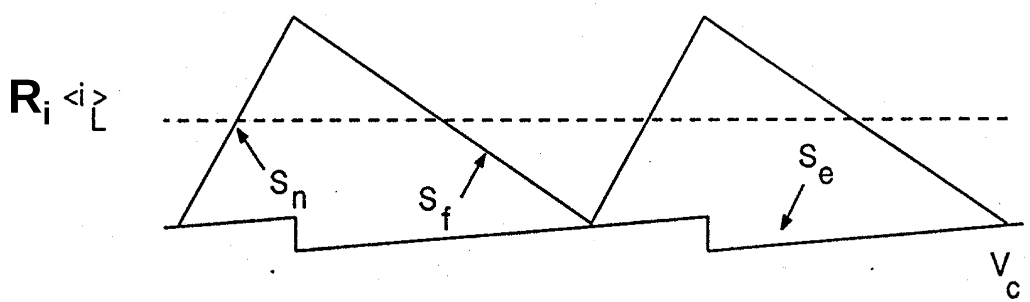
Circuit Waveforms
Find Average Inductor Current



Controlled on-time

(a)

$$R_i < i_L >= V_c - dT_sS_e - \frac{S_f d'T_s}{2}$$

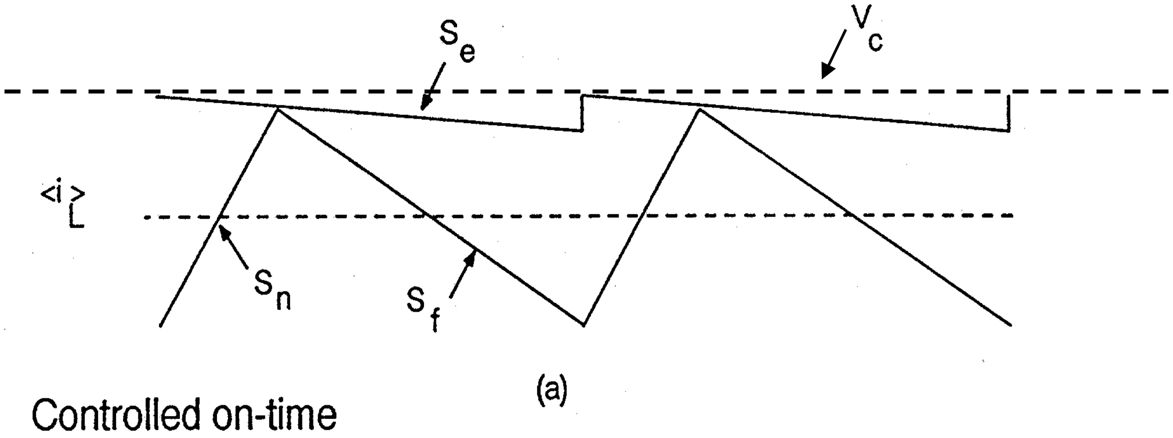


Controlled off-time

(b)

$$R_i < i_L >= V_c + d'T_sS_e + \frac{S_f dT_s}{2}$$

Use constant frequency, trailing-edge control as an example ……



$$R_i < i_L >= V_c - dT_sS_e - \frac{S_f d' T_s}{2}$$

$$d = \frac{v_{off}}{v_{off} + v_{on}} \qquad d' = \frac{v_{on}}{v_{off} + v_{on}} \qquad S_f = \frac{v_{off}R_i}{L}$$

Only perturbing v_{on}

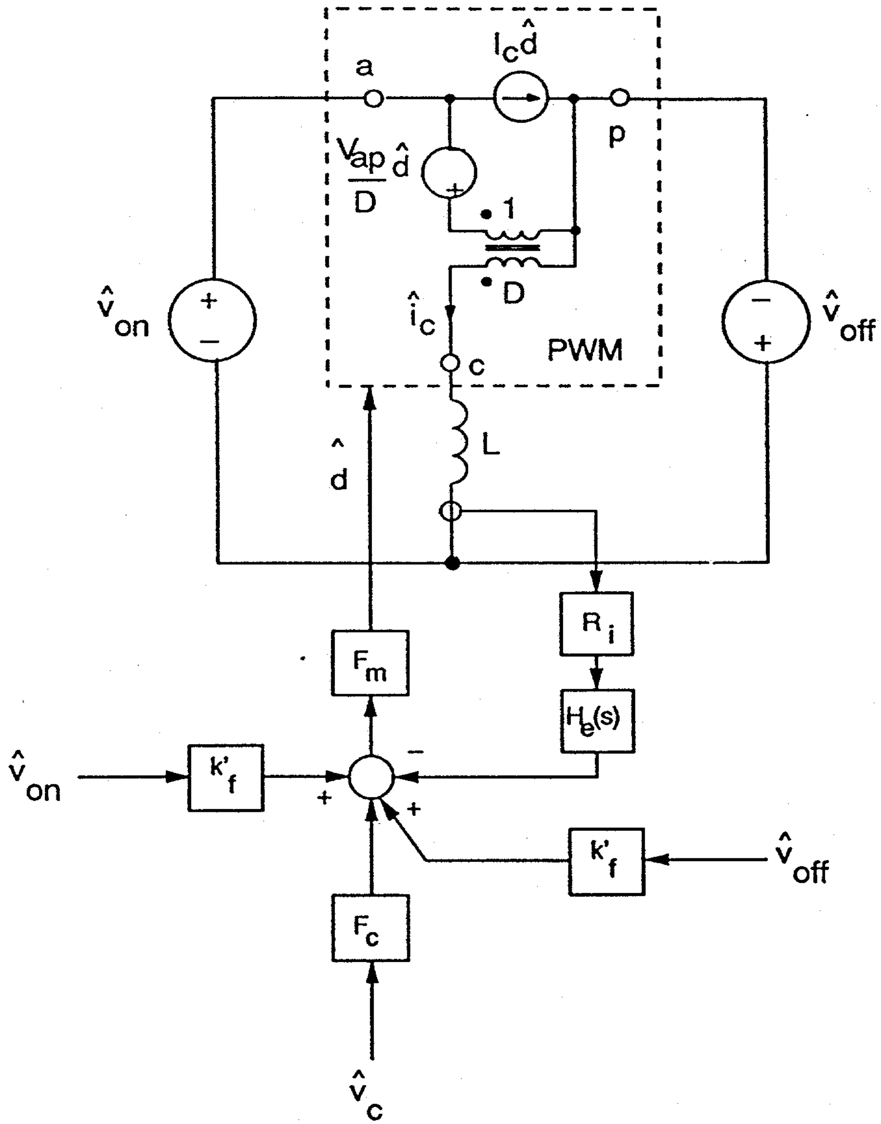
to get $\left\langle \hat{i}_L \right\rangle$ $\Rightarrow$

$$\boxed{\frac{\left\langle \hat{i}_L \right\rangle}{\hat{v}_{on}} = \frac{DS_eT_s}{V_{ap}R_i} - \frac{D^2T_s}{2L}}$$

… Equ. 1

Block Diagram

Find Average Inductor Current



- **Let:** $\hat{v}_{off} = 0 \quad \hat{v}_c = 0$

- **Notice:** at DC (or low frequency), He(s) =1, and the inductor is a short circuit (or very low impedance).

$$\Downarrow$$

$$\left(\hat{v}_{on} k_f{}' - \left\langle \hat{i}_L \right\rangle R_i\right) \cdot F_m \cdot \frac{V_{ap}}{D} = -\hat{v}_{on}$$

$$\Downarrow$$

$$\boxed{\frac{\left\langle \hat{i}_L \right\rangle}{\hat{v}_{on}} = \frac{1}{R_i}\left[\frac{D}{F_m V_{ap}} + k_f{}'\right]} \quad \text{… Equ. 2}$$

Equation 1 = Equation 2

$$\frac{\left\langle \hat{i}_L \right\rangle}{\hat{v}_{on}} = \frac{DS_eT_s}{V_{ap}R_i} - \frac{D^2T_s}{2L} \qquad \text{...... Equ. 1}$$

$$\frac{\left\langle \hat{i}_L \right\rangle}{\hat{v}_{on}} = \frac{1}{R_i}\left[\frac{D}{F_mV_{ap}} + k_f'\right] \qquad \text{...... Equ. 2}$$

$$\text{(and:} \quad F_m = \frac{1}{(S_n + S_e)T_s}$$

for constant - freqency, trailing - edge control)

$$\Rightarrow \qquad \boxed{k_f' = \frac{-DT_sR_i}{L}\left[1 - \frac{D}{2}\right]}$$

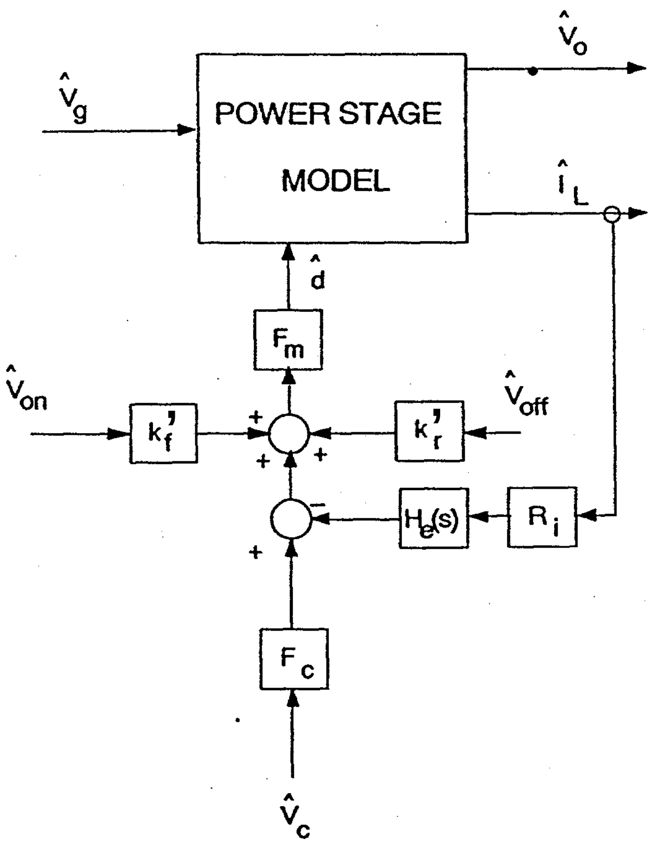
Similarly: Equation 1' = Equation 2'

$$\frac{\left\langle \hat{i}_L \right\rangle}{\hat{v}_{off}} = \frac{-D' S_e T_s}{V_{ap} R_i} - \frac{D'^2 T_s}{2L} \quad \text{...... Equ. 1'}$$

$$\frac{\left\langle \hat{i}_L \right\rangle}{\hat{v}_{off}} = \frac{1}{R_i}\left[\frac{-D'}{F_m V_{ap}} - k_r'\right] \quad \text{...... Equ. 2'}$$

$$\Rightarrow \quad \boxed{k_r' = \frac{D'^2 T_s R_i}{2L}}$$

Complete Current-Mode Model

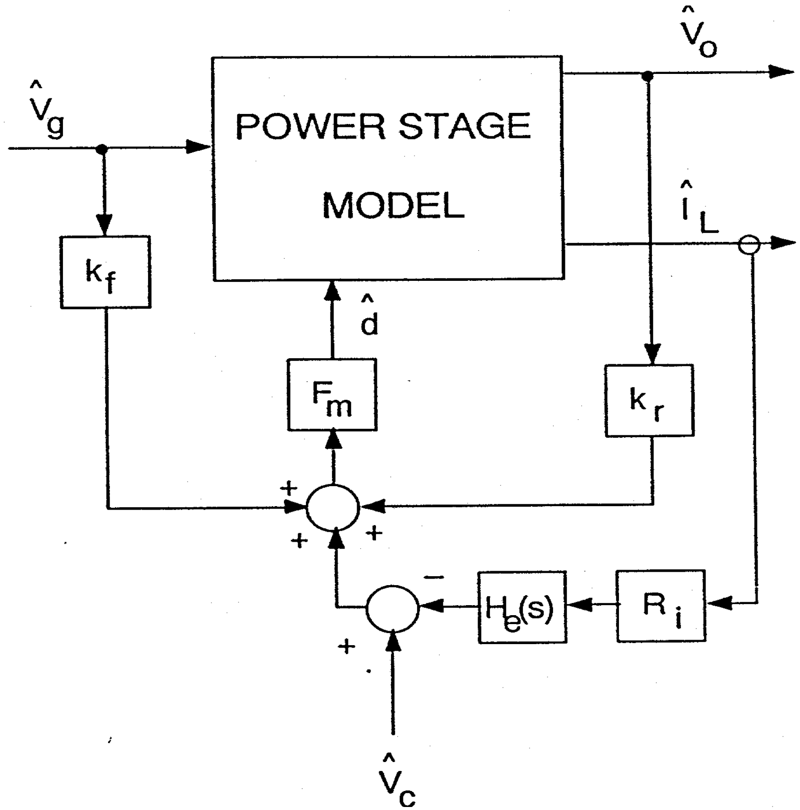


peak current mode valley current mode

	Constant Frequency Trailing Edge	Constant Frequency Leading Edge	Constant Off-Time	Constant On-Time
k_f'	$-\frac{DT_sR_i}{L}\left[1-\frac{D}{2}\right]$	$-\frac{D^2T_sR_i}{2L}$	$-\frac{DT_sR_i}{L}$	$-\frac{DT_sR_i}{2L}$
k_r'	$\frac{D'^2T_sR_i}{2L}$	$\frac{D'T_sR_i}{L}\left[1-\frac{D'}{2}\right]$	$\frac{D'T_sR_i}{2L}$	$\frac{D'T_sR_i}{L}$
F_m	$\frac{1}{(S_n+S_e)T_s}$	$\frac{1}{(S_f+S_e)T_s}$	$\frac{D'}{S_nT_s}$	$\frac{D}{S_fT_s}$
F_c	1	1	$e^{sDT_s/2}$	$e^{sD'T_s/2}$
$H_e(s)$	$1+\frac{s}{\omega_nQ_z}+\frac{s^2}{\omega_n^2}$		$Q_z=\frac{-2}{\pi}$	$\omega_n=\frac{\pi}{T_s}$

variable freq control, a Fc term is to match the simulation

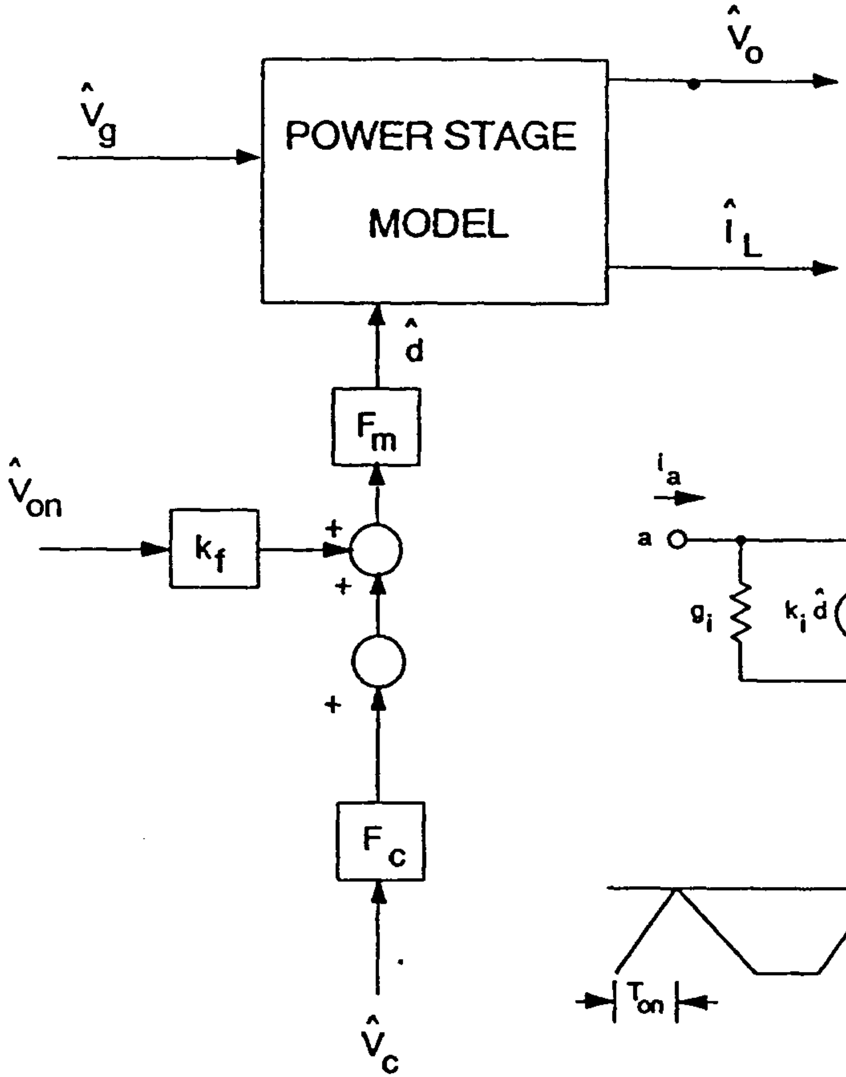
Alternate Form of Model



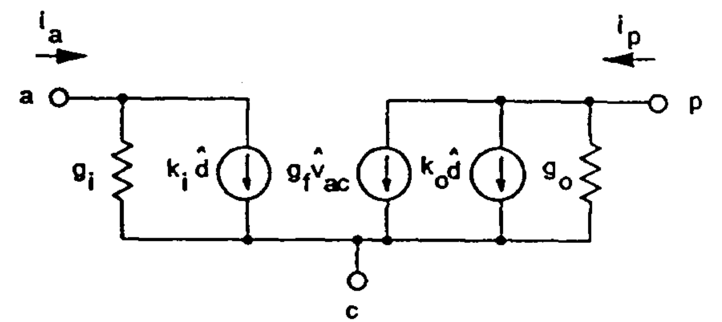
- **Feedforward terms connected to input and output voltages**
- **This form of the model can be more useful for analytical derivations of transfer functions**

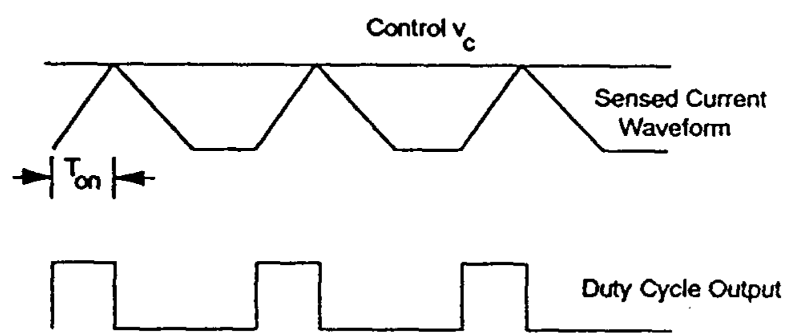
	Buck	Boost	Buck-Boost
k_f	$-\frac{DT_s R_i}{L}\left[1-\frac{D}{2}\right]$	$-\frac{T_s R_i}{2L}$	$-\frac{DT_s R_i}{L}\left[1-\frac{D}{2}\right]$
k_r	$\frac{T_s R_i}{2L}$	$\frac{D'^2 T_s R_i}{2L}$	$\frac{D'^2 T_s R_i}{2L}$

Complete Current-Mode Model (DCM)



if converter works in DCM pk current mode, the current sense is no more important, low controllability in current loop



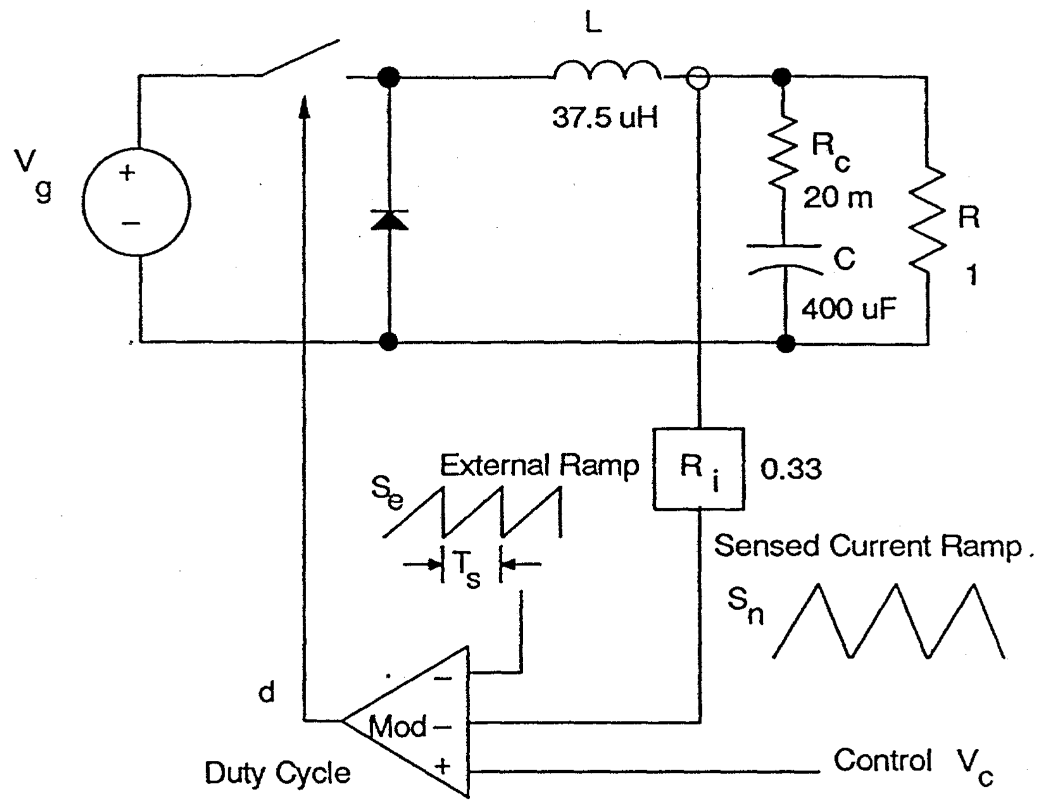


Complete Current-Mode Model (DCM)

	Constant Frequency	Constant Off-Time
F_m	$\frac{1}{(S_n + S_e)T_s}$	$\frac{D'}{S_n T_s}$
F_c	1	$e^{sDT_s/2}$
k_f	$-\frac{DT_s R_i}{L}$	

- **DCM model does not require any sampled-data modeling**

Example of Model for Buck Converter



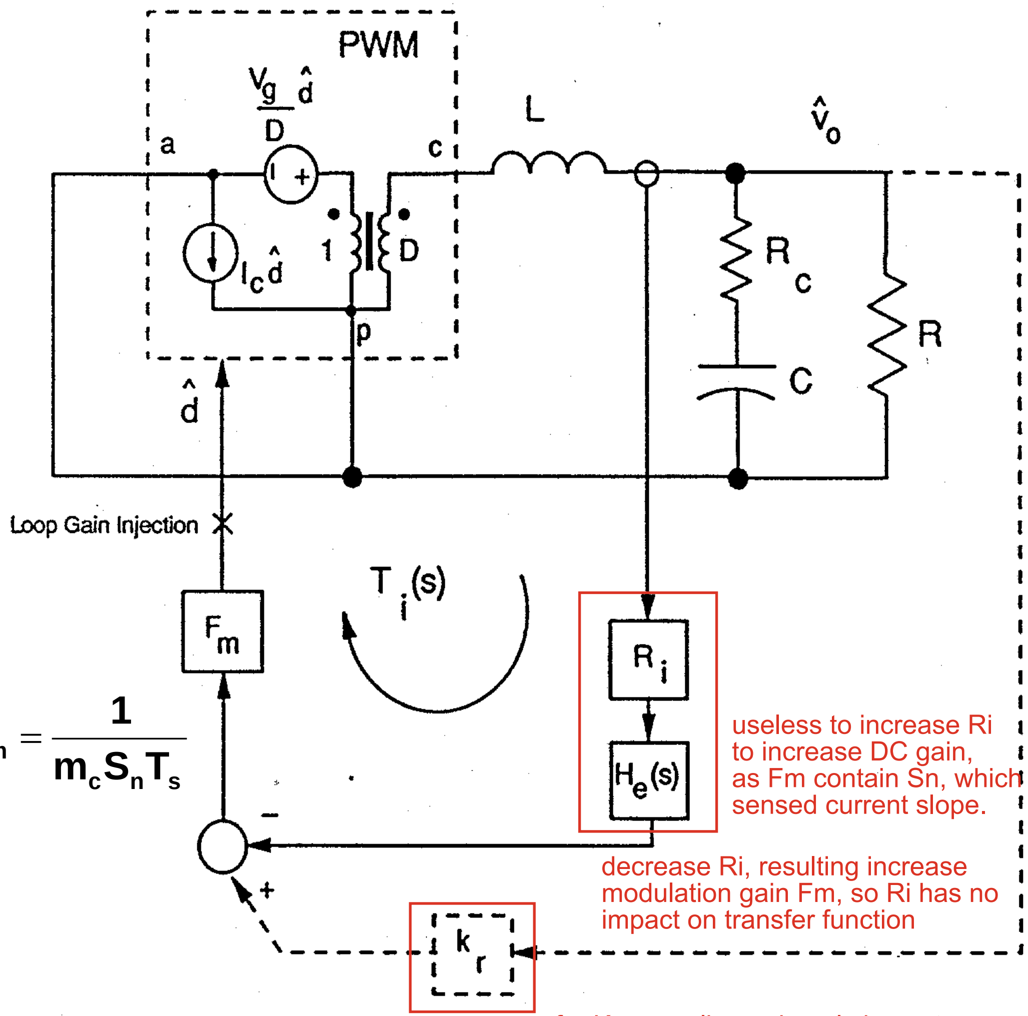
$T_s = 20\mu s$

$V_o = 5V$

$V_g = 11V$

- **Buck converter shows some of the most interesting properties**

Current-Loop Gain of the Buck



$$F_m = \frac{1}{m_c S_n T_s}$$

useless to increase Ri to increase DC gain, as Fm contain Sn, which is sensed current slope.

decrease Ri, resulting increase modulation gain Fm, so Ri has no impact on transfer function

same for Kr, sampling gain only impact low freq part, as Vo is average value and ignored the high freq ripples

DC gain (mc is the only control para)

$$T_i(s) \approx \boxed{\frac{L}{RT_s m_c D'}} \frac{1+sCR_c}{\Delta(s)} H_e(s)$$

$$\Delta(s) = 1 + \frac{s}{\omega_0 Q_{ps}} + \left(\frac{s}{\omega_0}\right)^2$$

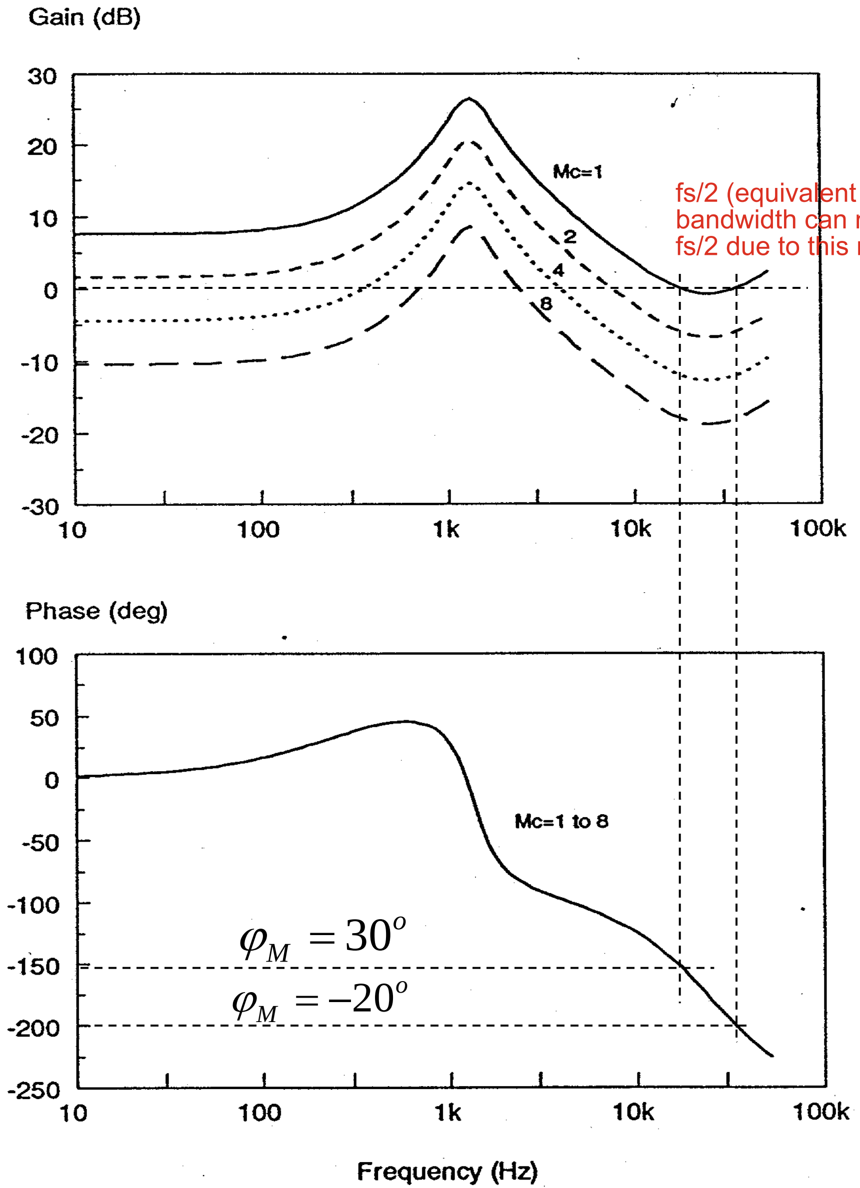
$$Q_{ps} = \frac{1}{\omega_0\left[\frac{L}{R} + CR_c\right]}$$

$$\omega_0 = \frac{1}{\sqrt{LC}}$$

$$H_e(s) \approx 1 + \frac{s}{\omega_n Q_z} + \frac{s^2}{\omega_n^2}$$

$$\omega_n = \frac{\pi}{T_s} \qquad Q_z = \frac{-2}{\pi}$$

Current-Loop Gain T_i



fs/2 (equivalent double zero) bandwidth can not larger then fs/2 due to this reason

$$m_c = 1 + \frac{S_e}{S_n}$$

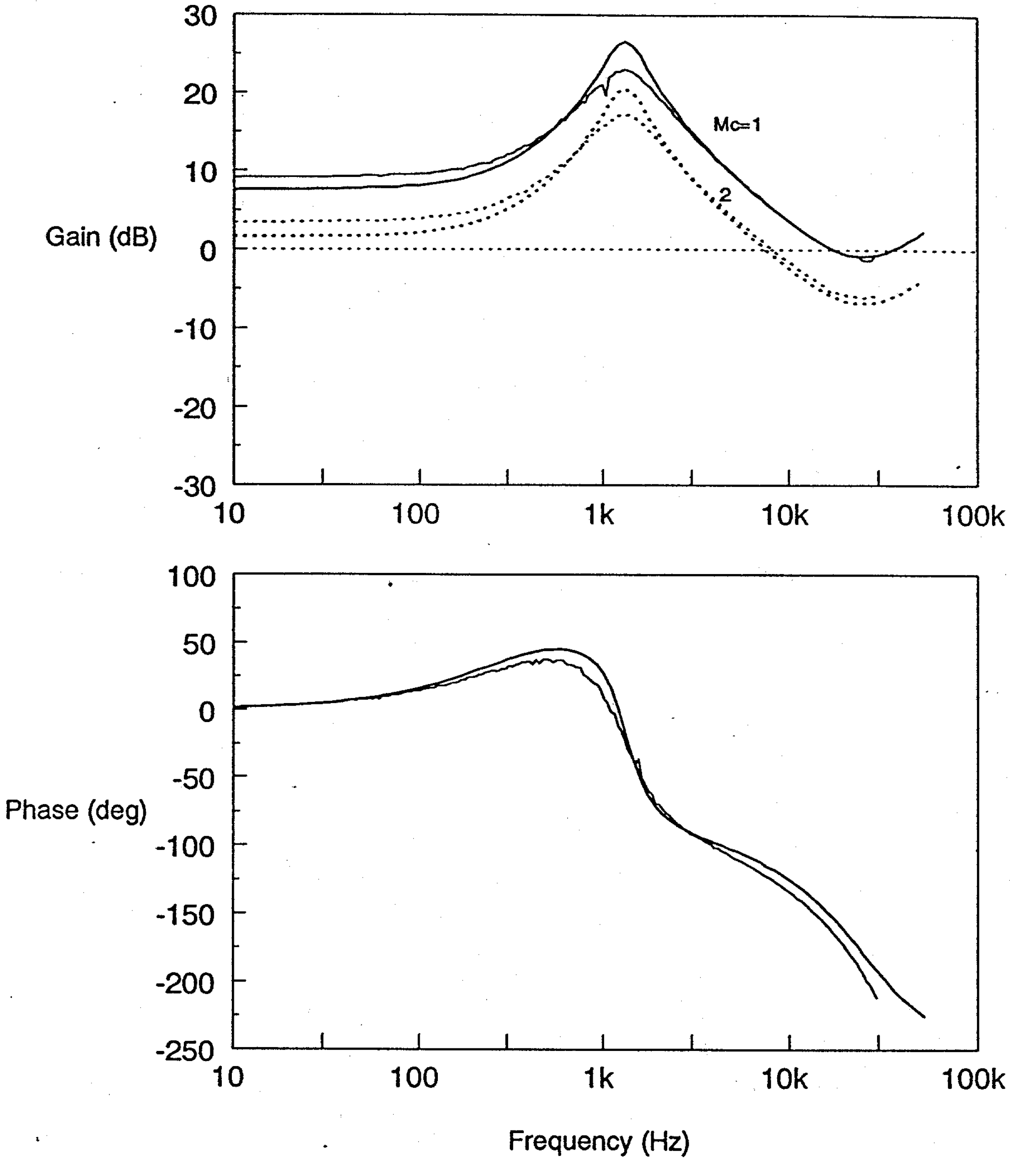
comp slope

leading slope

- External ramp will impact gain of T_i
- Reason for instability clearly shown

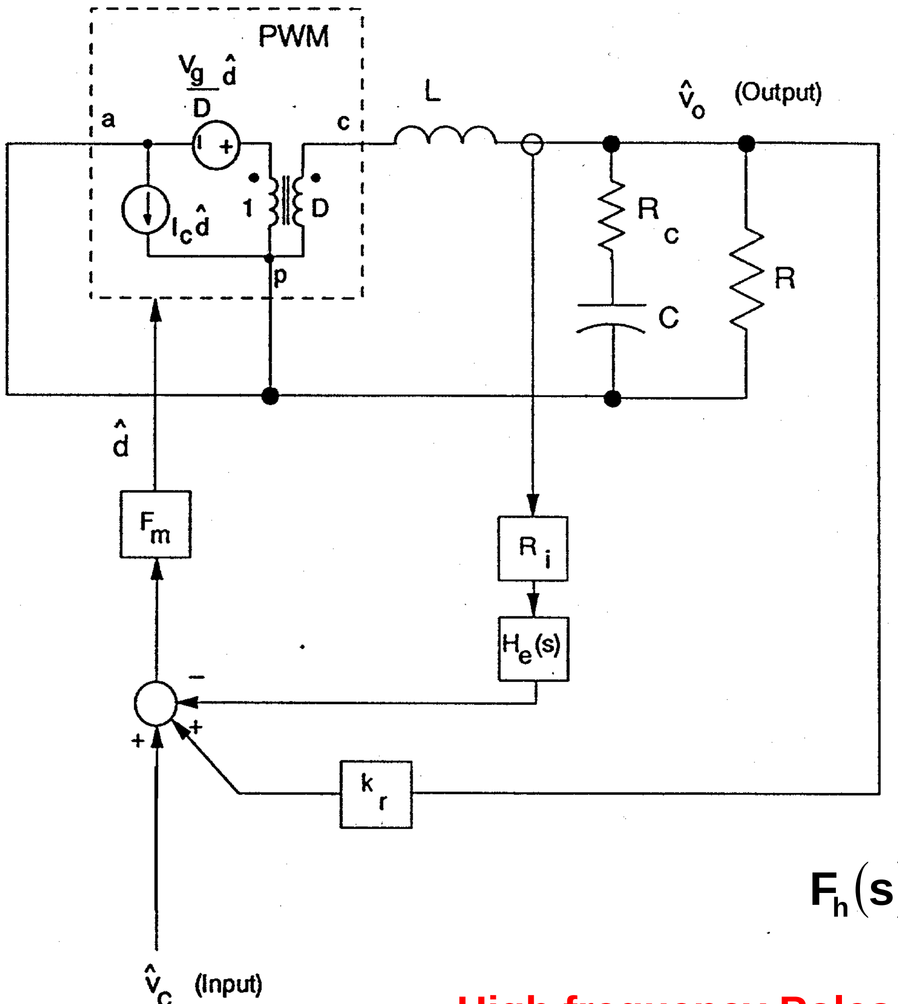
Buck Current-Loop Gain

Experimental Results



- **Experimental and theoretical results agree closely**
- **Addition of small external ramp stabilizes system**

Control-to-Output Transfer Function



- **Current-loop is closed**

$$\frac{\hat{v}_o}{\hat{v}_c} \approx \frac{R}{R_i}\frac{1}{1+\frac{RT_s}{L}\left[m_cD' - 0.5\right]}F_p(s)F_h(s)$$

$$F_p(s) = \frac{1+sCR_c}{1+\frac{s}{\omega_p}}$$

Low frequency Pole due to power stage

$$\omega_p = \boxed{\frac{1}{CR}} + \frac{T_s}{LC}\left(m_cD' - 0.5\right)$$

dominant

inductor is no longer free to resonant with capacitor, it is more likely to a current source

$$F_h(s) = \frac{1}{1+\frac{s}{\omega_n Q_p}+\left(\frac{s}{\omega_n}\right)^2}$$

$$Q_p = \frac{1}{\pi\left(m_cD' - 0.5\right)}$$

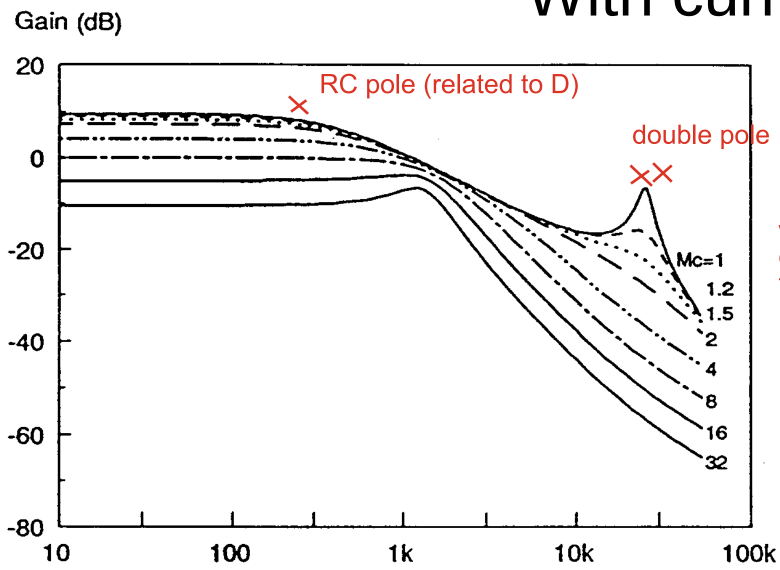
when Q is negative, double pole move to RHP, means unstable in closed loop system

$$\omega_n = \frac{\pi}{T_s}$$

High frequency Poles due to sampling effect

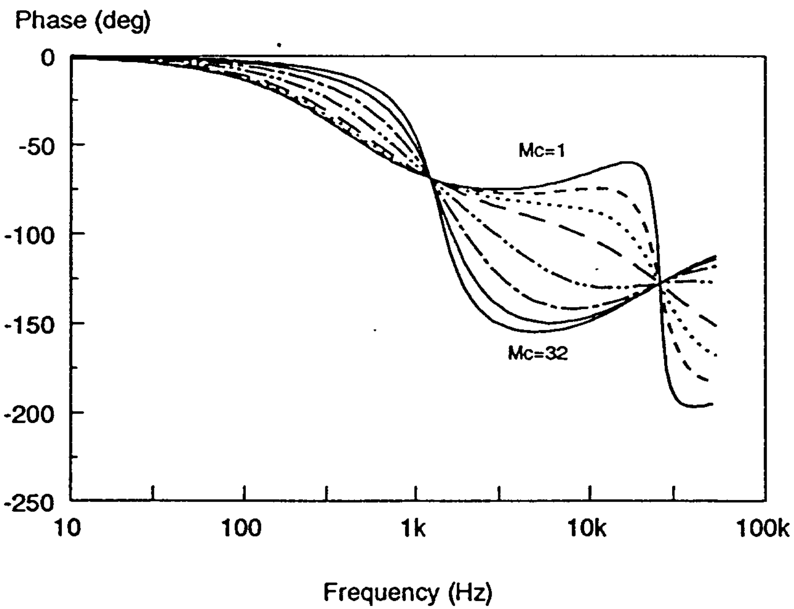
Control-to-Output Predictions

With current-loop closed



when mc increase, the peak gets shallow and eventually disappear
double pole split and one go to low freq, one go to high freq (ignored)
the low freq pole may meet with RC pole and generate a new double pole (open loop vin->vo double pole)

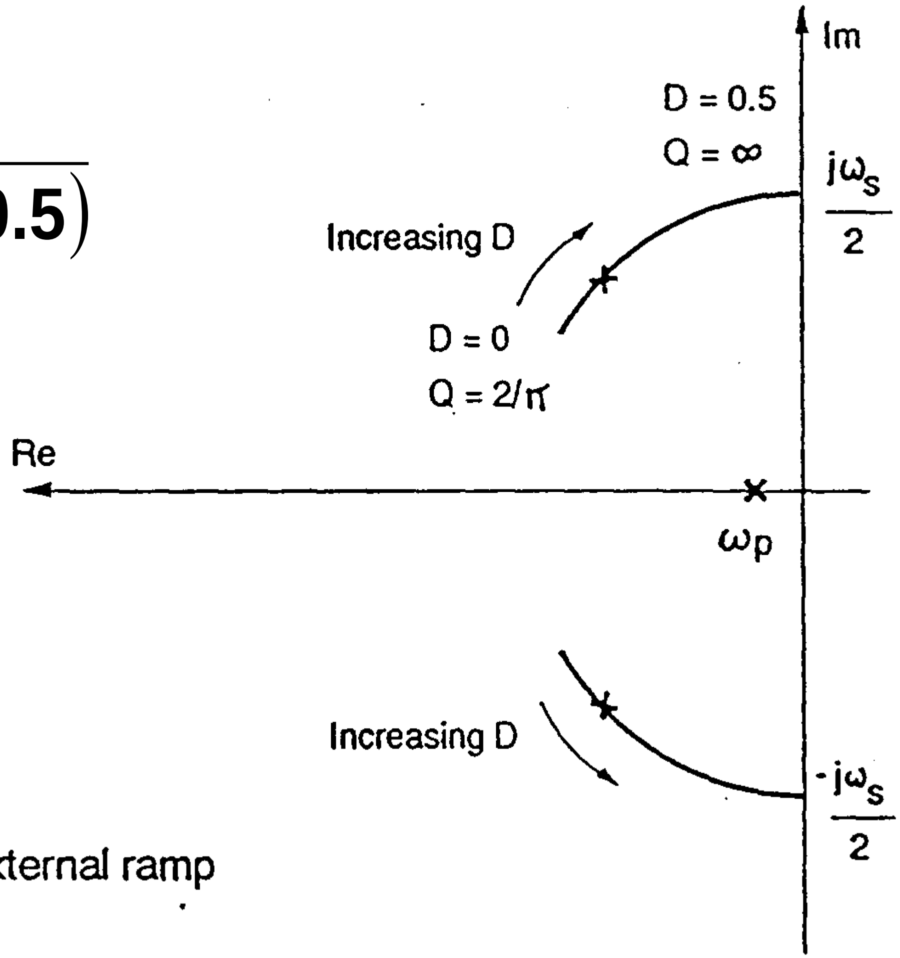
- **Double pole at half the switching frequency**
- **Become voltage-mode characteristics with large external ramp**



chose a good mc to have double pole but no peak
(a small Q but double pole is not split)
1. avoid double cross over
2. have less phase reduce in medium freq range

Poles of System with Current-Loop Closed

$$Q_p = \frac{1}{\pi\left(m_c D' - 0.5\right)}$$

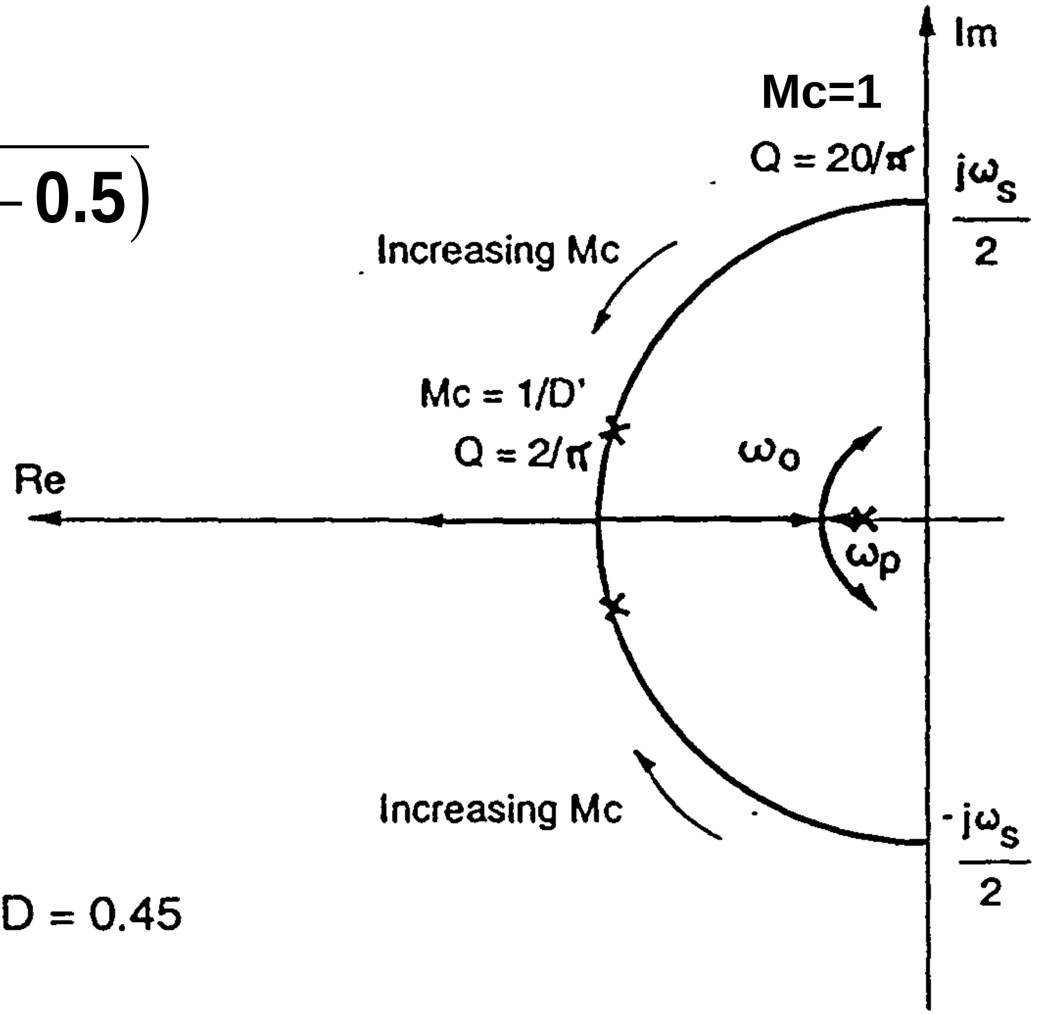


(a) No external ramp

$$m_c = 1 + \frac{S_e}{S_n} = 1$$

Poles of System with Current-Loop Closed

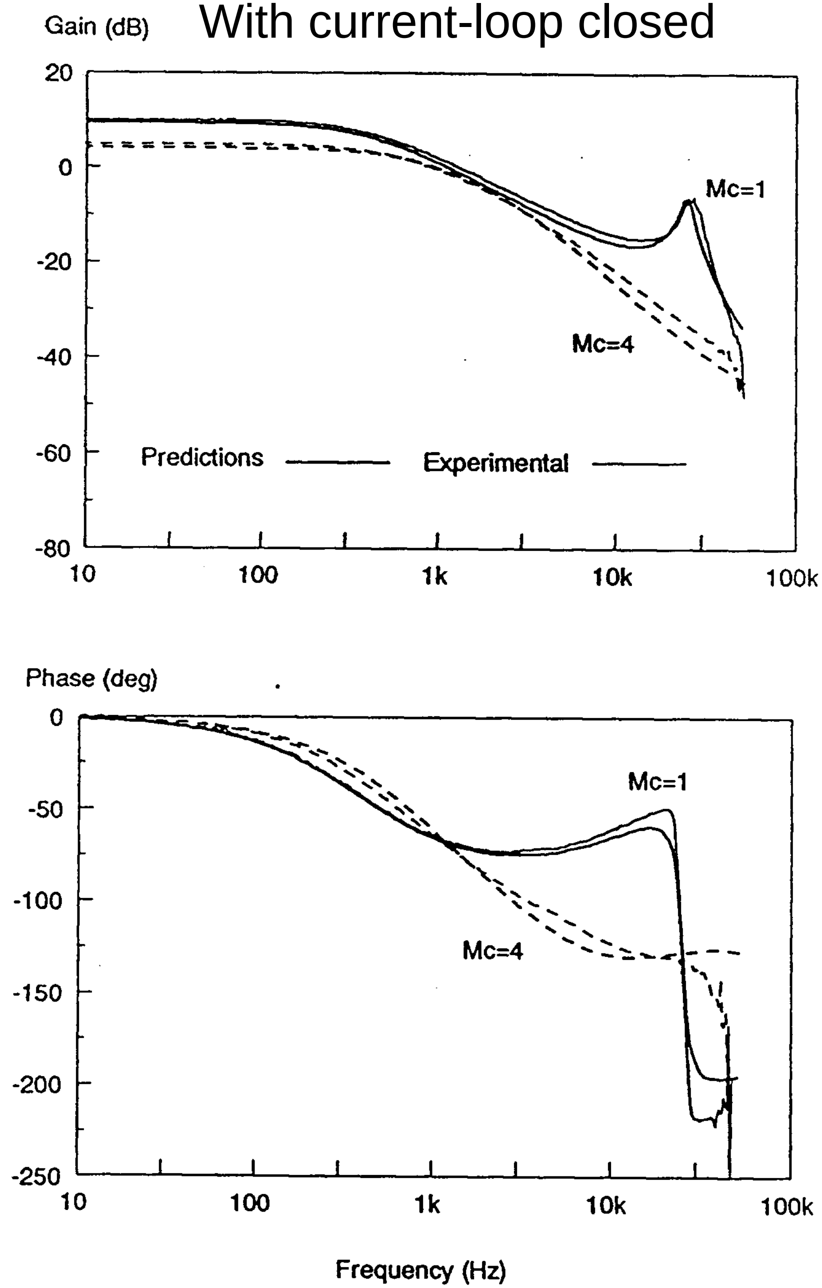
$$Q_p = \frac{1}{\pi\left(m_c D' - 0.5\right)}$$



(b) Fixed D = 0.45

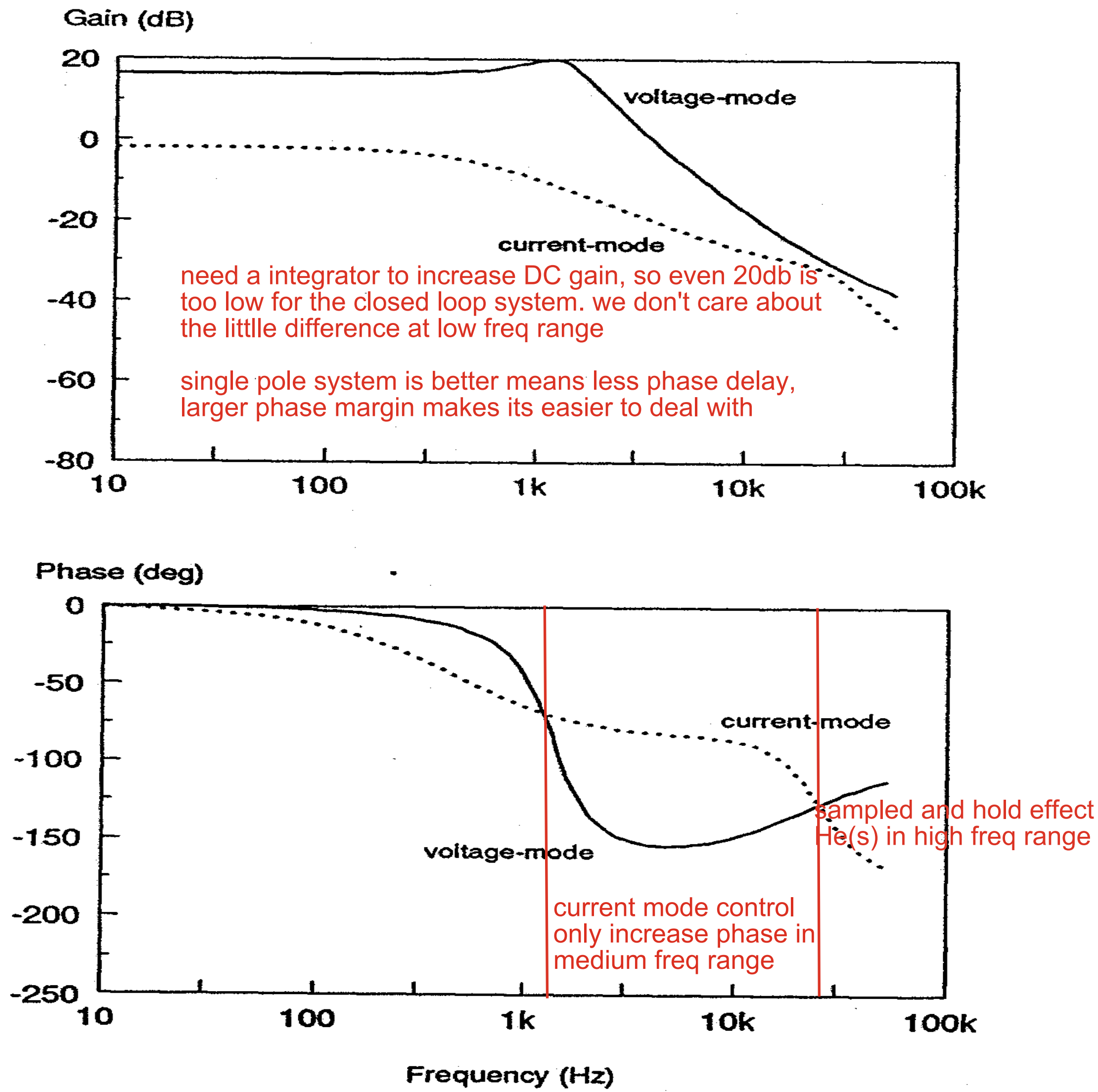
Control-to-Output Measurement

With current-loop closed

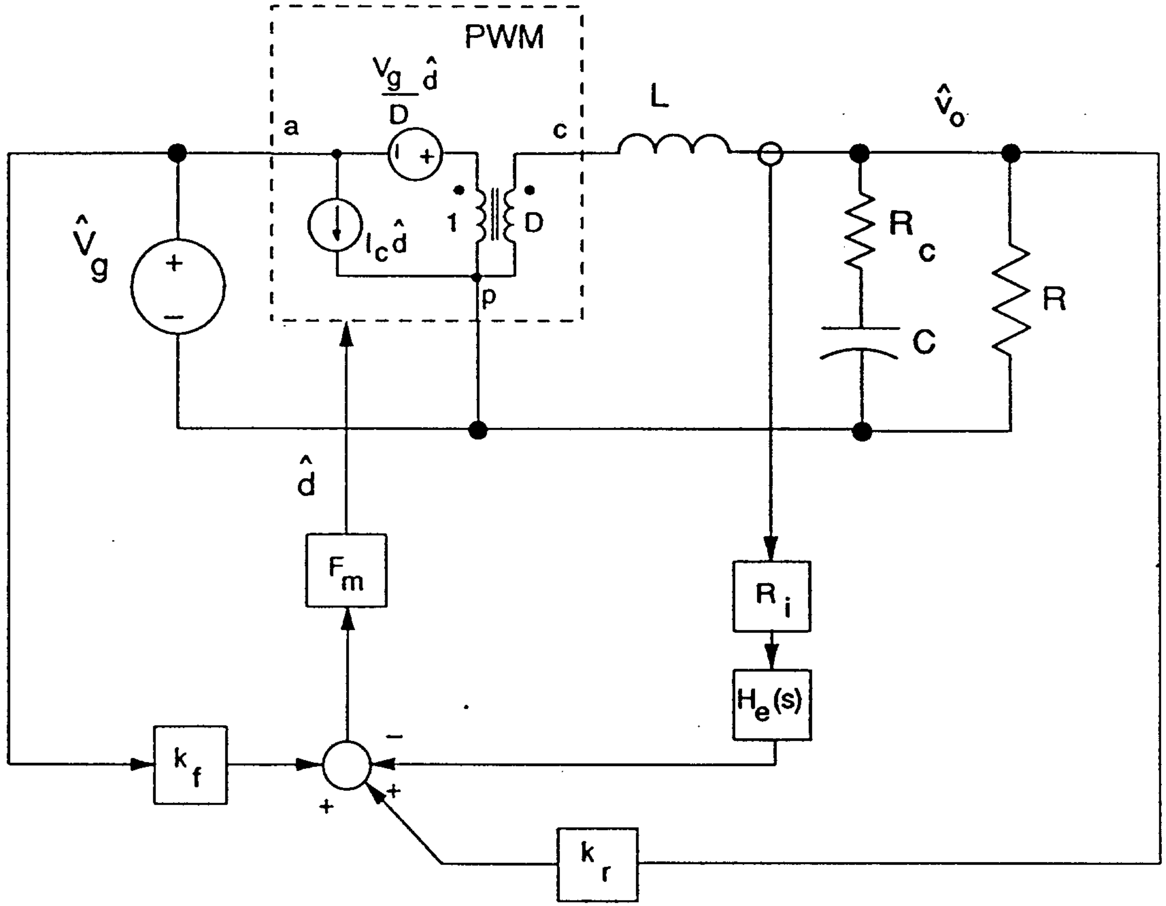


Effect of Current-Mode Control
Control-to-Output

- **Resonant frequency eliminated**
- **Double poles at half switching frequency**

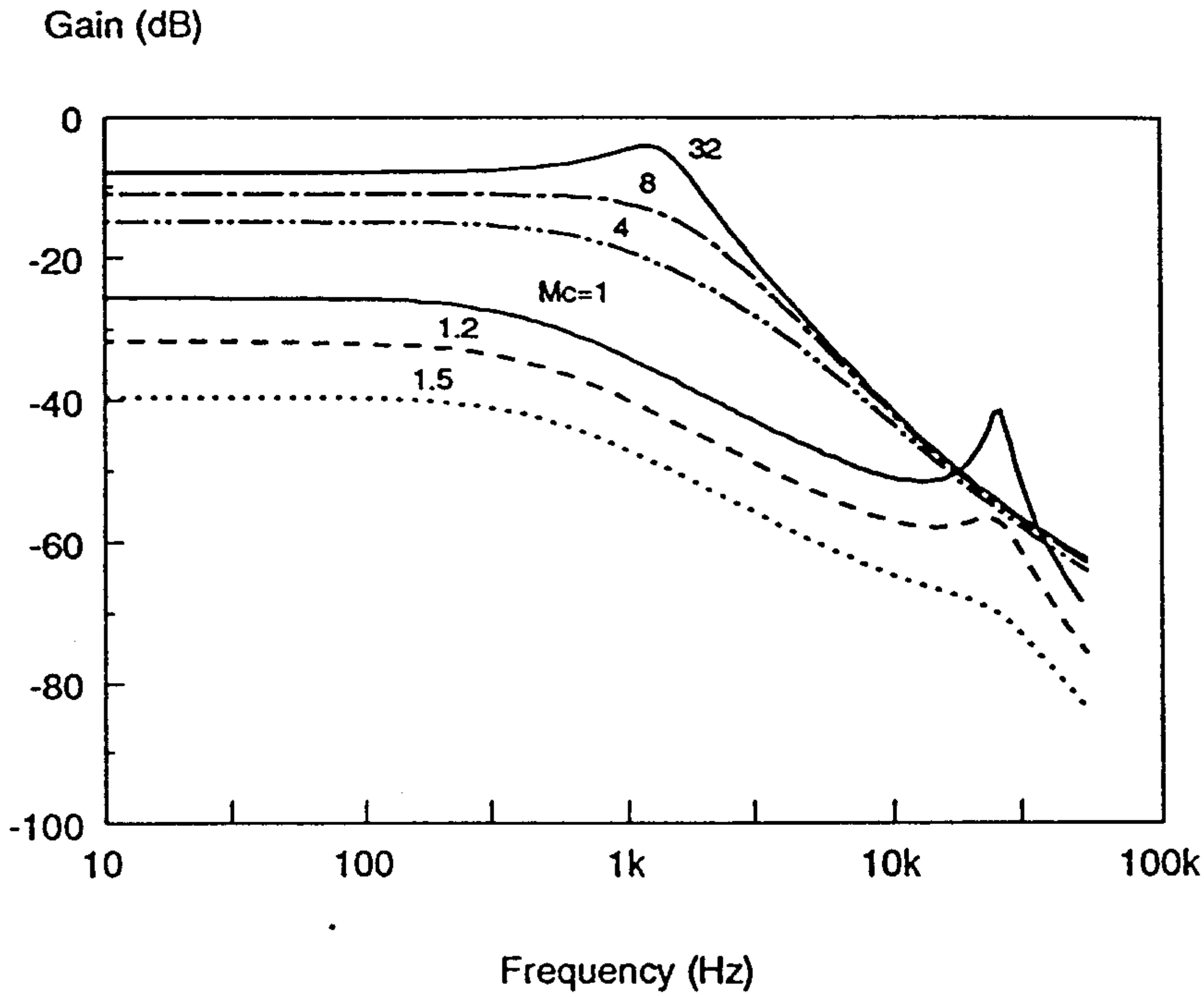


Audio Susceptibility Transfer Function

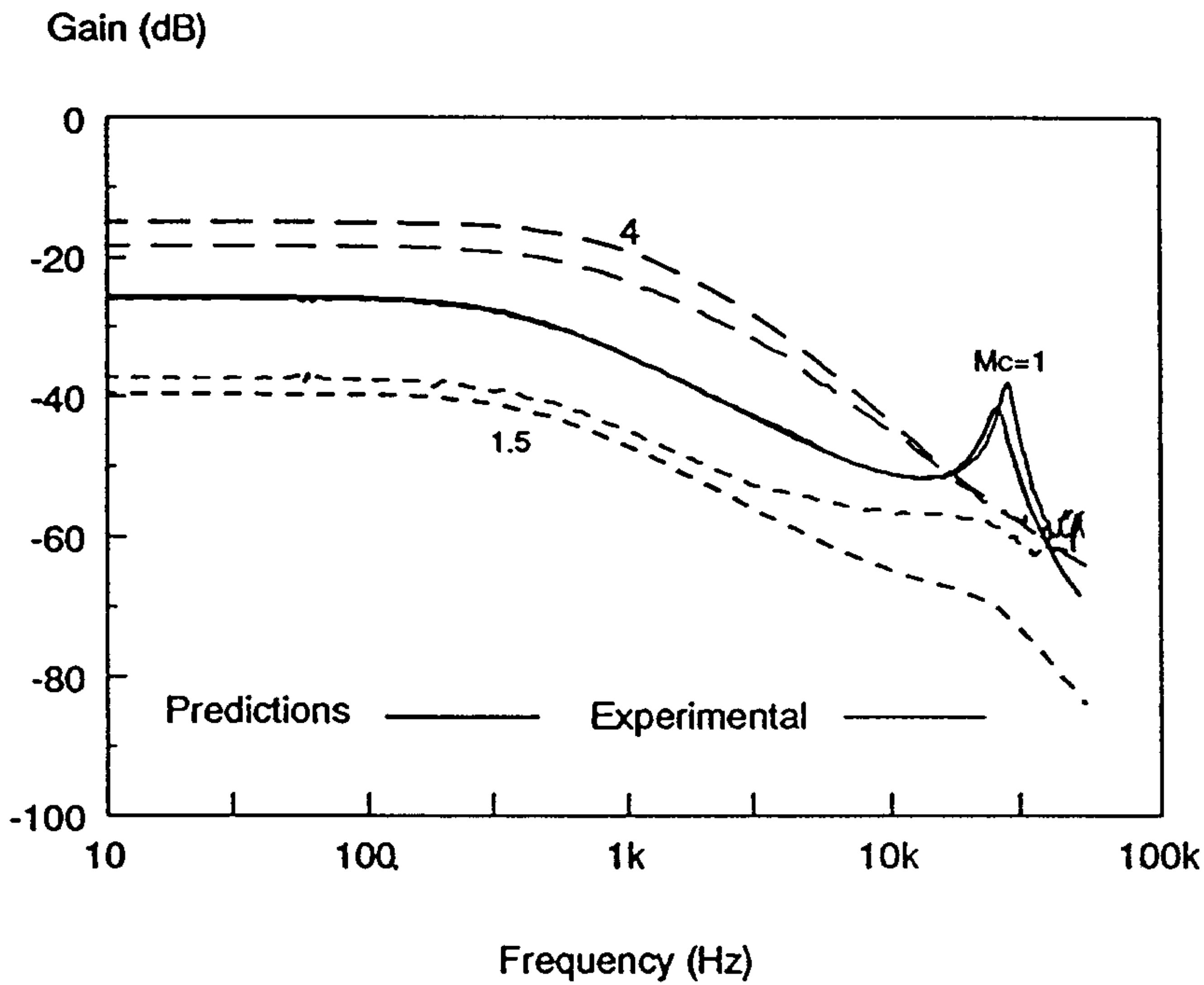


$$\frac{\hat{v}_o}{\hat{v}_g} = \frac{D\left[m_c D' - (1 - D/2)\right]}{\frac{L}{RT_s} + (m_c D' - 0.5)} F_p(s) F_h(s)$$

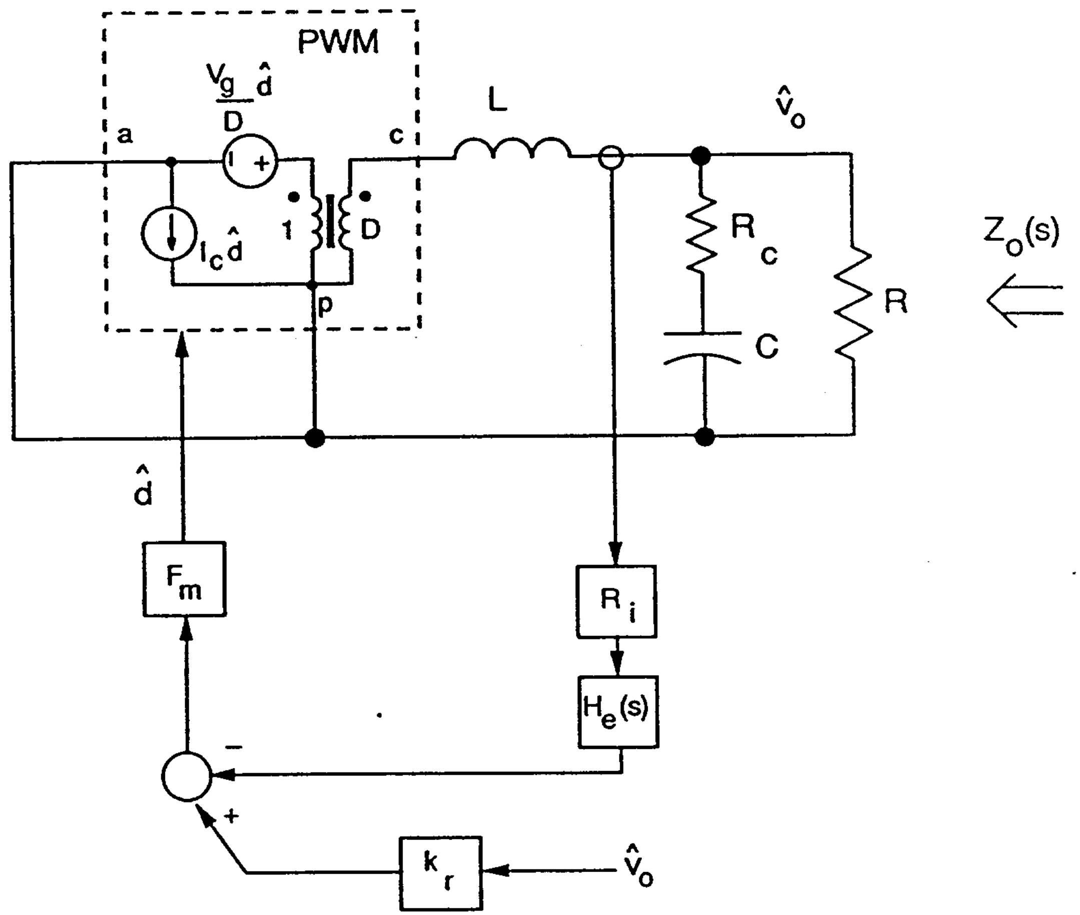
Audio Susceptibility Predictions



Audio Susceptibility Measurement

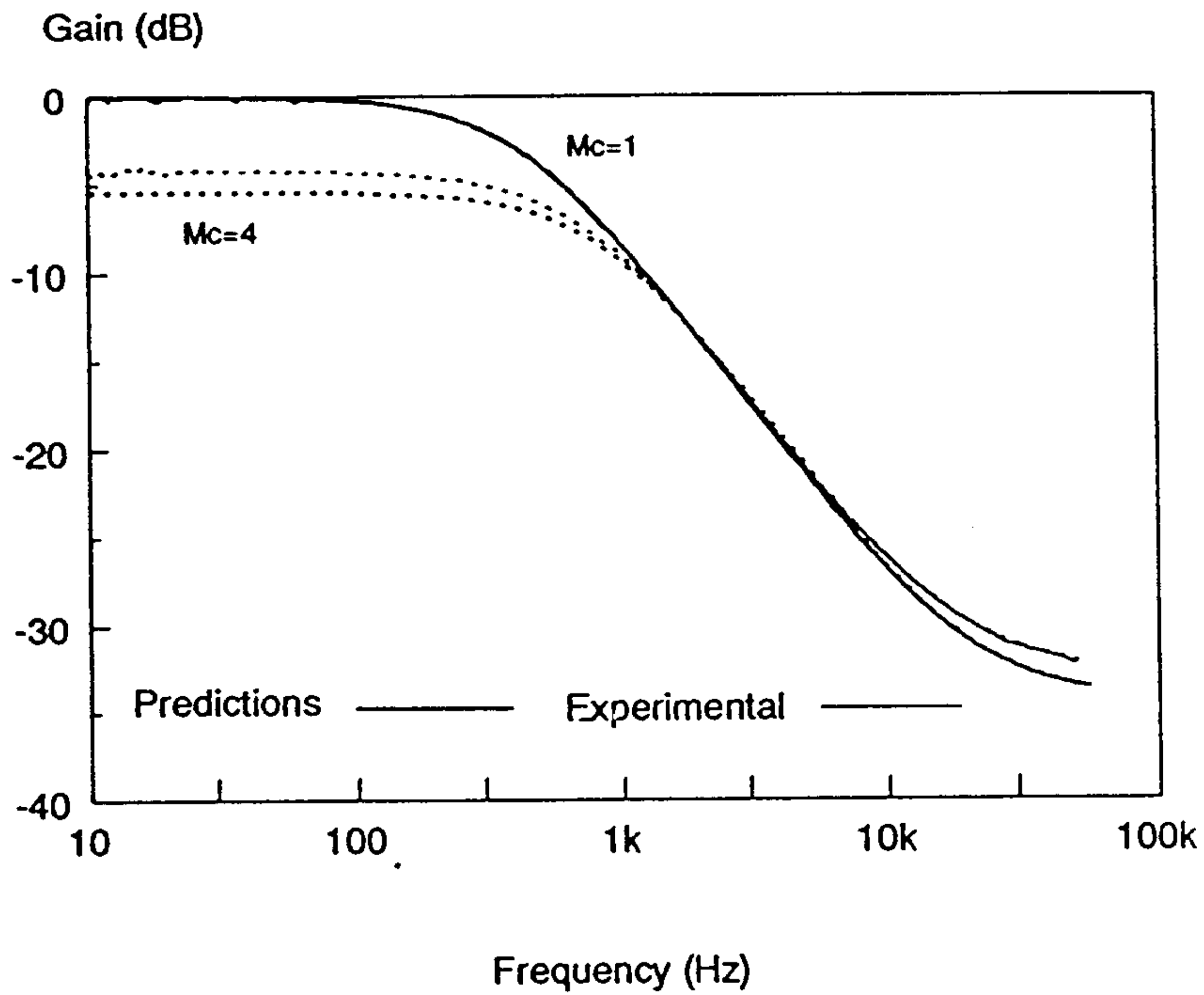


Output-Impedance Transfer Function



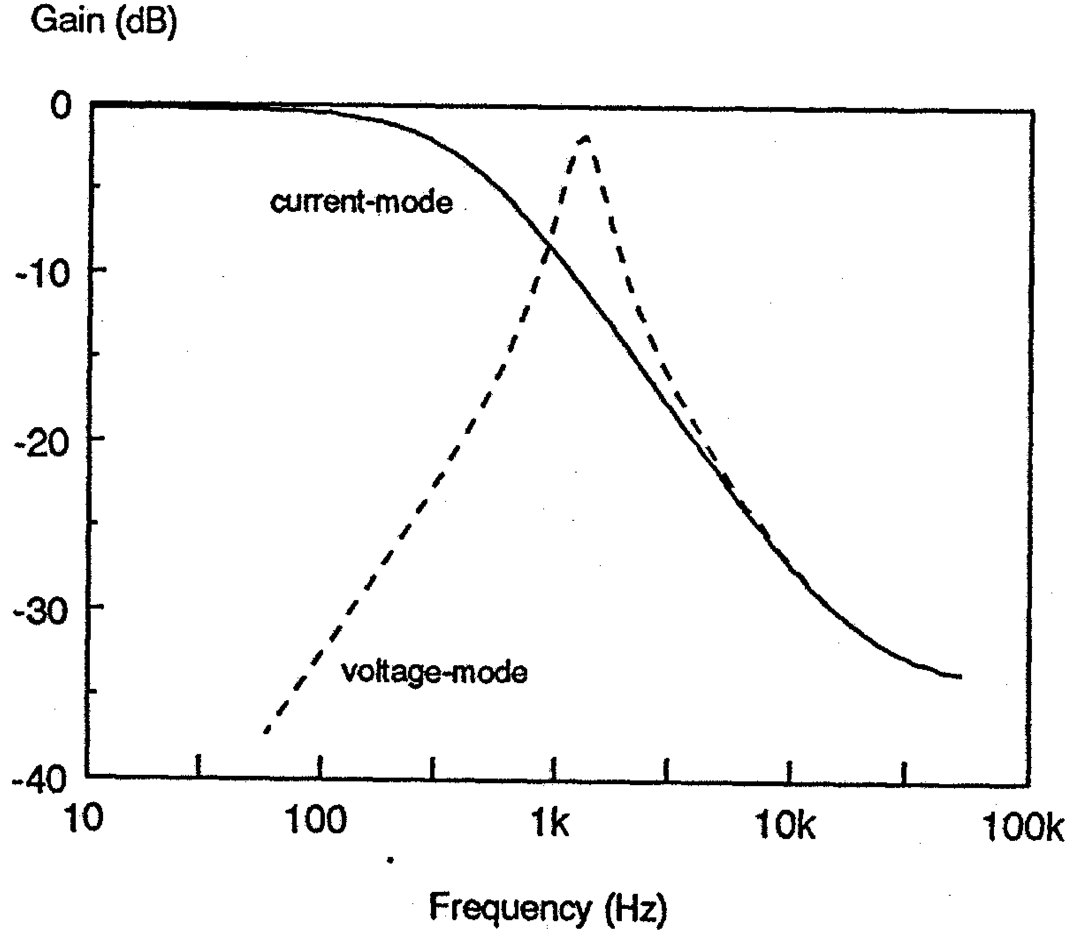
$$Z(s) \approx \frac{R}{1 + \frac{RT_s}{L}\left(m_c D' - 0.5\right)} F_p(s)$$

Output Impedance Measurement



Effect of Current-Mode Control
Output Impedance

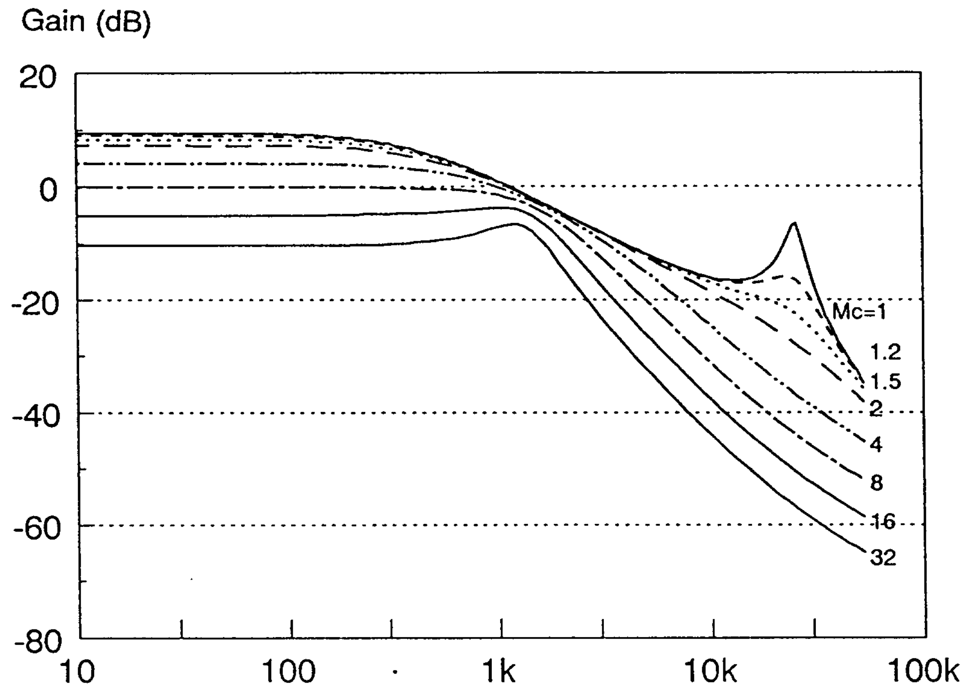
- **High output impedance at frequencies below resonant frequency due to "current source"**
- **Open-loop dc regulation is poor**



Design of Peak-Current-Mode Control Systems

Select the External Ramp

- **Select the external ramp to damp the complex double poles at half the switching frequency**



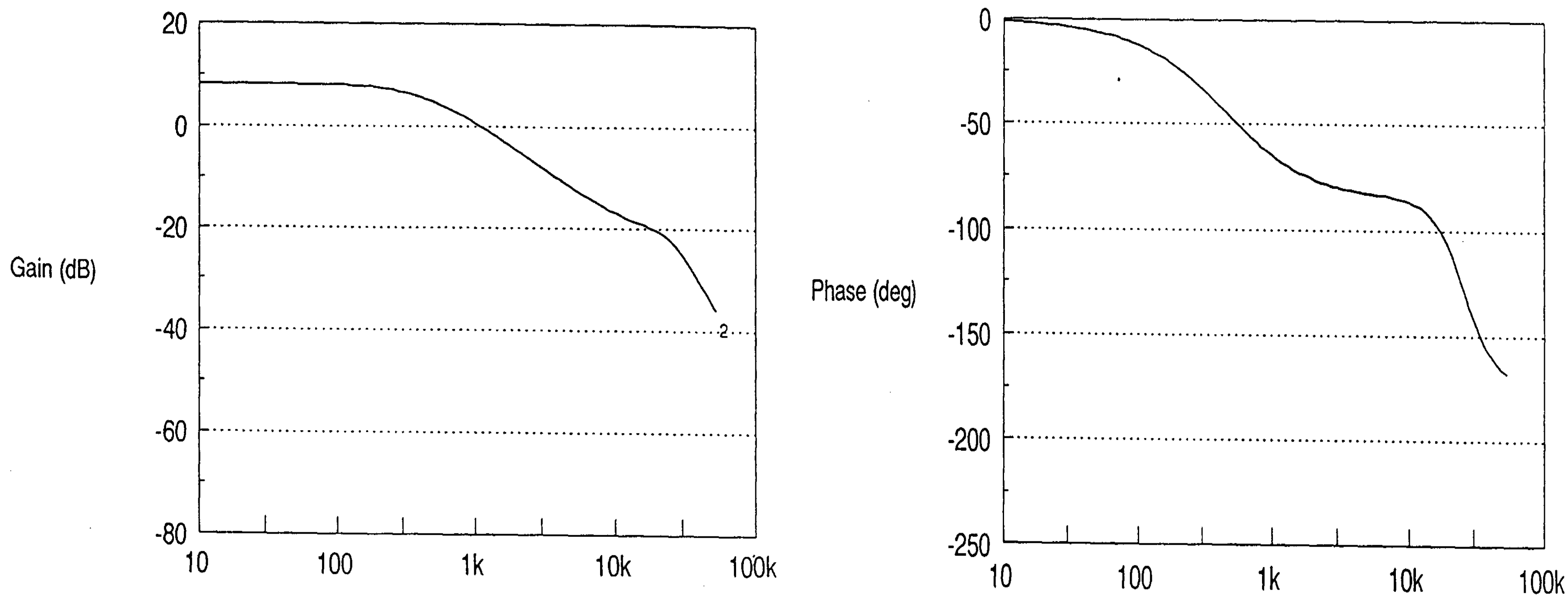
- **Damping of the double pole is given by:**

$$Q = \frac{1}{\pi(m_C D' - 0.5)}$$

- **For a Q of 1, select external ramp to be**

$$S_e = V_{g_{min}} \frac{(D_{max} - 0.18)}{\tau_m} \qquad \tau_m = \frac{L}{R_i} \qquad m_c = 1 + \frac{S_e}{S_n} = 1$$

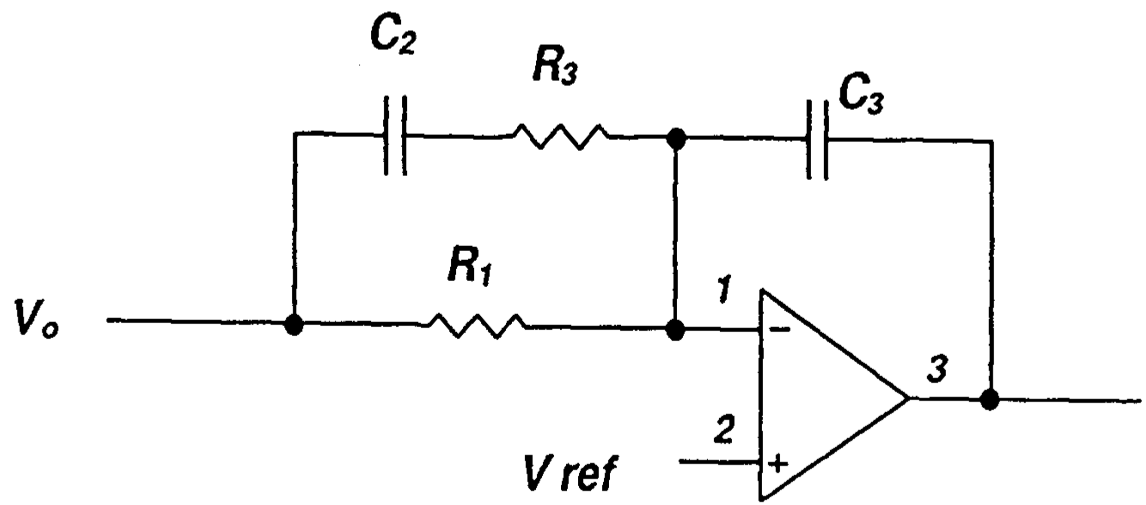
Measure the Control-Output Transfer Function



- **Low frequency pole**
- **Double pole at half switching frequency**
- **ESR zero and RHP zeros same as voltage mode**

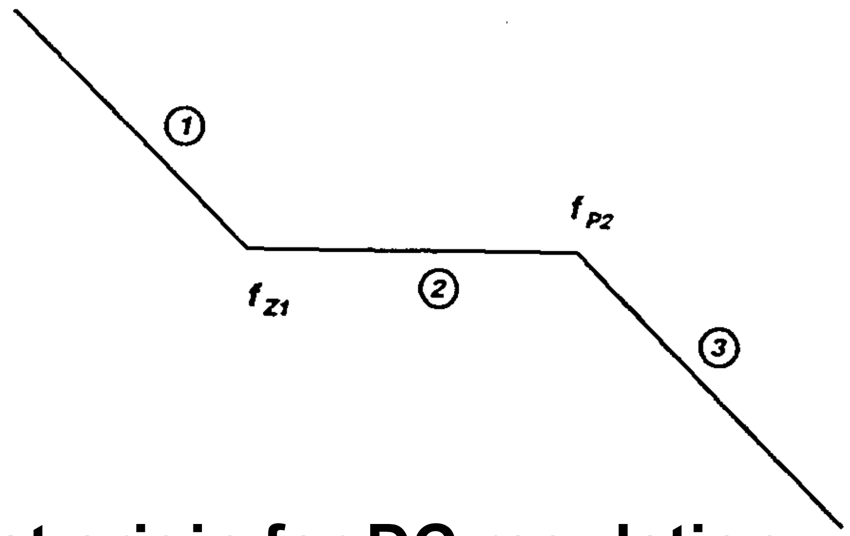
Design Voltage-Loop Compensation

Compensation network for current-mode control



- Simple two-pole, single zero network is adequate
- Design is much easier than for voltage-mode control compensation

Compensation asymptotes for current-mode control



- Pole at origin for DC regulation
- Zero placed for desired settling time constant
- Second pole placed at power stage ESR zero, RHP zero, or half the switching frequency, whichever is lower.
- Gain adjusted for desired phase margin